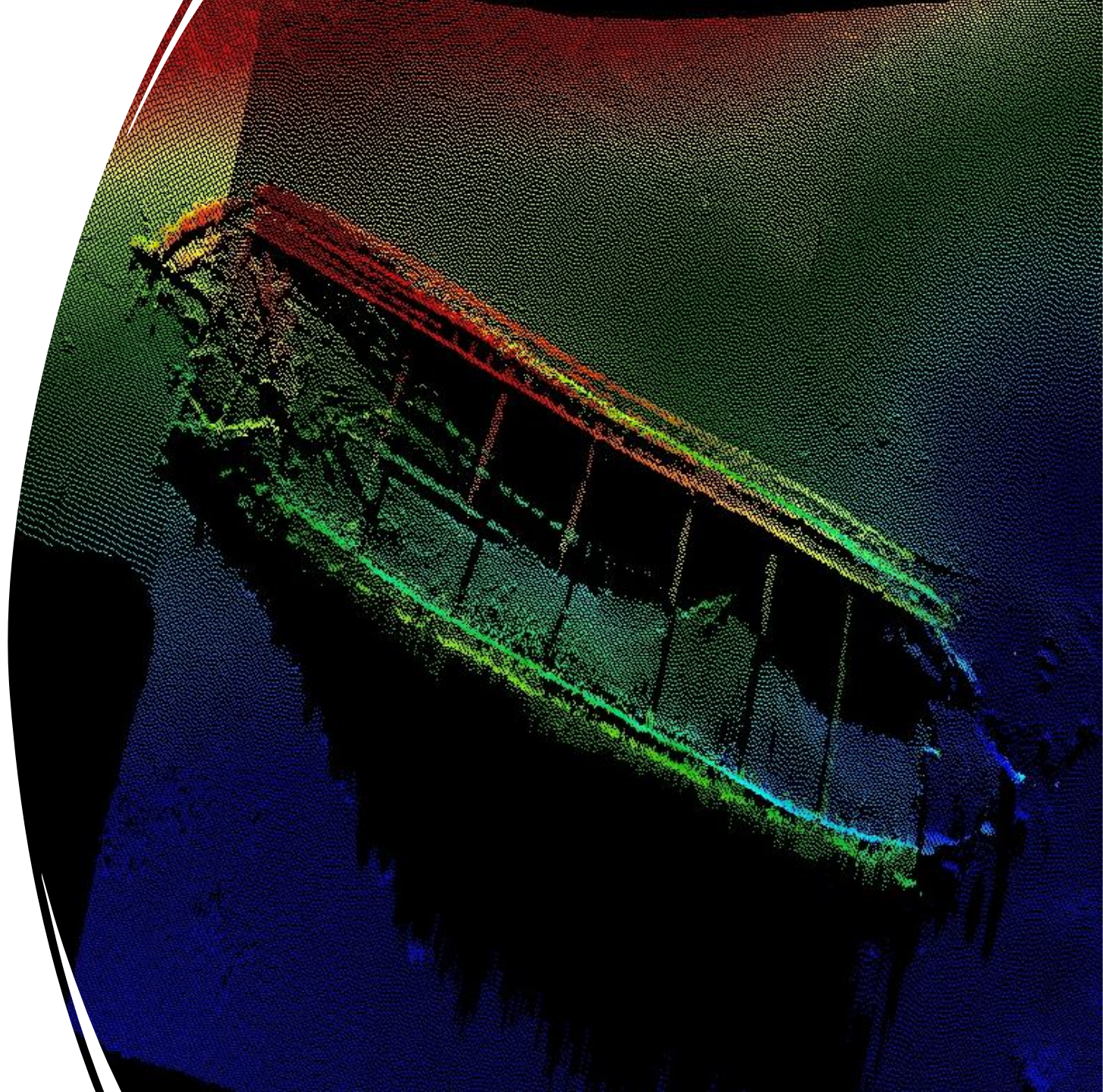


HYDROGRAPHIC SURVEYING FOR LAND SURVEYORS

Jared Lambert & Gabriela De Oliveira
February 19, 2026

COURSE OUTLINE

- Introduction to Hydrographic surveying
- McKim & Creed Overview
- Foundational Concepts
- Key Tools and Technologies
- Key Applications and Examples
- Bathymetric Survey Methods
- Accuracy Standards
- System Checks and Calibrations
- Brief Overview of Geophysical Surveys
- Project Workflow
- Project Considerations
- Mission Planning
- Data Processing and Integration
- Final Products
- Education Opportunities
- Q&A





About Jared

- Hydrographic Survey Director at McKim & Creed (14 Years)
- AAS in Marine Technology (2005) 20+ yrs in the Industry
- NC A&T Geomatics Program (current)
- Passionate about my Family and Outdoors



About Gabi

- Passionate about the Ocean
- SCUBA Diving Instructor
- Undergrad and grad in Biological Sciences
- Marine Mammal Observer
- Seismic Navigator
- AAS in Marine Technology
- Hydrographic Production Manager
- Geomatics at NC A & T (2026)

ABOUT MCKIM & CREED



McKim & Creed combines
a **LOCAL TEAM** with a
DEEP BENCH of
NATIONAL RESOURCES
to ensure that
appropriate resources
and staff are allocated to
each project.



47

Years In
Business

36

National Offices

1000

+

U.S. Employees

#21

Southeast Top
Engineer Firms [ENR]

#141

Top 200 Environmental
Firms [ENR]

#167

Top 500
Design Firms [ENR]

OUR SERVICES



WATER / WASTEWATER

SCADA
Wastewater
Reclaimed Water
Water Management
Renewal & Replacement
Instrumentation & Controls
Trenchless Technology
Hydraulic Modeling
Alternative Delivery



MECHANICAL, ELECTRICAL, PLUMBING, FIRE PROTECTION

Mechanical
Electrical
Plumbing
Fire Protection
Commissioning
Solar



GEOMATICS

Subsurface Utility Engineering
GIS
Airborne LiDAR
Mobile LiDAR
Land Survey
3D Laser Scanning
Hydrographic Survey
Route Survey
Boundary Retracement



SITE / CIVIL

Master Planning
Landscape Architecture
Site/Civil Engineering
Stormwater
Low Impact Development (LID)
Roadway Design

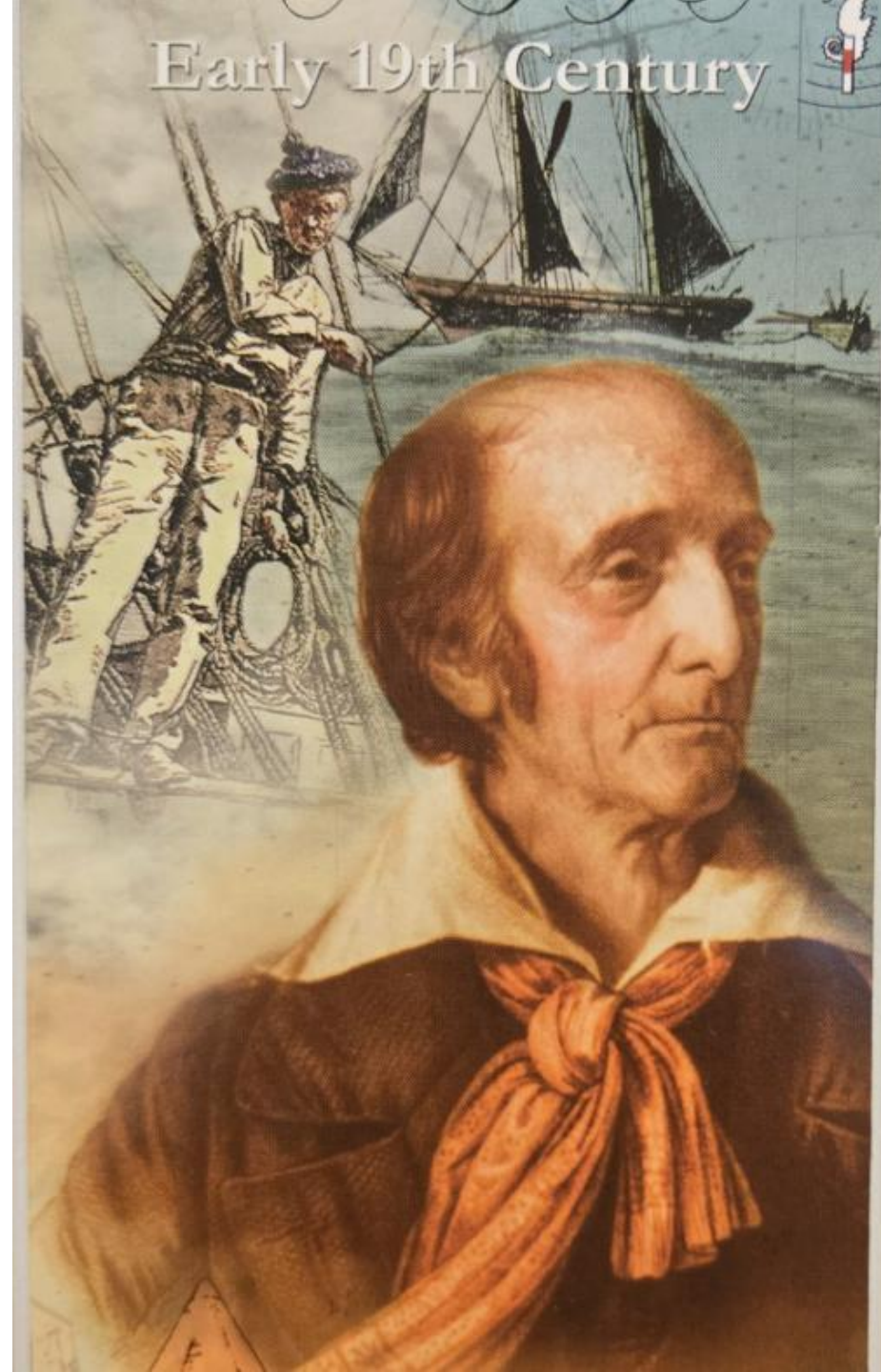


INDUSTRIAL DESIGN- BUILD

Project Development Design
Estimating
Procurement
Scheduling
Commissioning
Project Management
Construction

U.S. HYDROGRAPHY HISTORY

- Swiss American Scientist Ferdinand Hassler
 - Selected to head a new organization (1807) – The United States Survey of the Coast
 - The Survey's mission expanded from the 1807 goal of providing accurate nautical charts for the safety of maritime commerce to include most of the physical sciences including hydrography, geodesy, astronomy, topography, oceanography, tide and current measurement and prediction, seismology, magnetics, national standards, photogrammetry, and more.
 - First official Hydrographic (sounding) operation in 1835 aboard the schooner **Experiment**
 - Lead line and fix positions using sextants to chart the coast



FOUNDATIONAL CONCEPTS

BATHYMETRY

Bathymetry is the study of the "beds" or "floors" of water bodies, including the ocean, rivers, streams, and lakes.

The term "bathymetry" originally referred to the ocean's depth relative to sea level, although it has come to mean "submarine topography," or the depths and shapes of underwater terrain.

In the same way that topographic maps represent the three dimensional features (or relief) of overland terrain, bathymetric maps illustrate the land that lies underwater. Variations in sea floor relief may be depicted by color and contour lines called depth contours or isobaths.

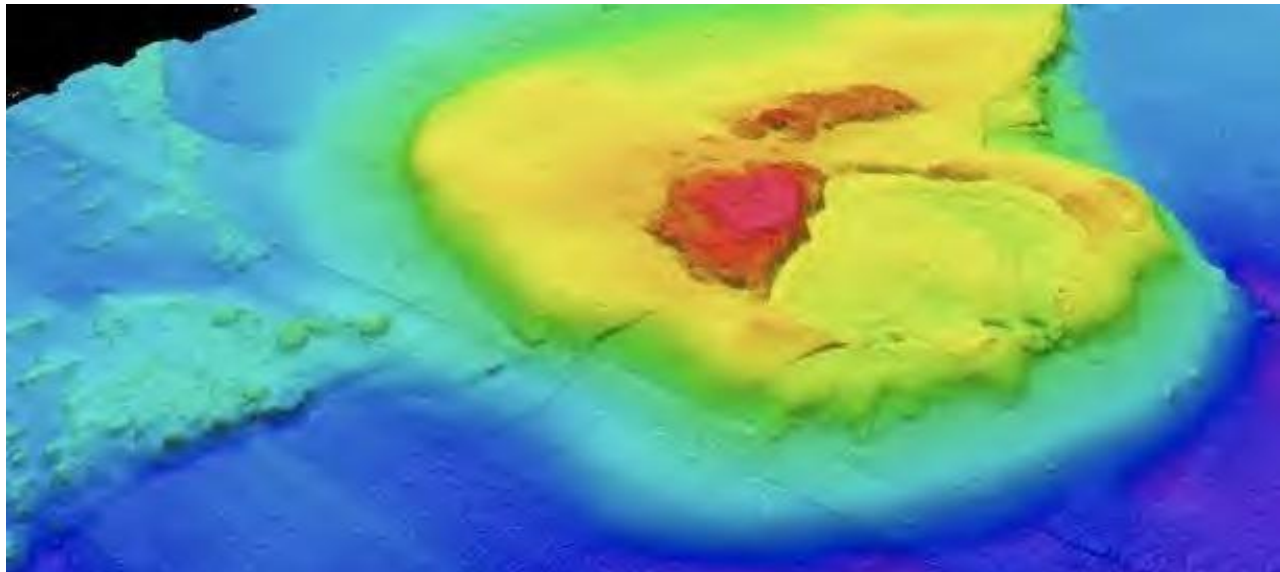
Bathymetry is the foundation of the science of hydrography.

HYDROGRAPHY

Hydrography is the science that measures and describes the physical features of navigable portion of the Earth's surface, bodies of water and the land areas adjacent to those bodies of water.

Hydrographic surveyors study these bodies of water to see what the seafloor/floor/bottom looks like.

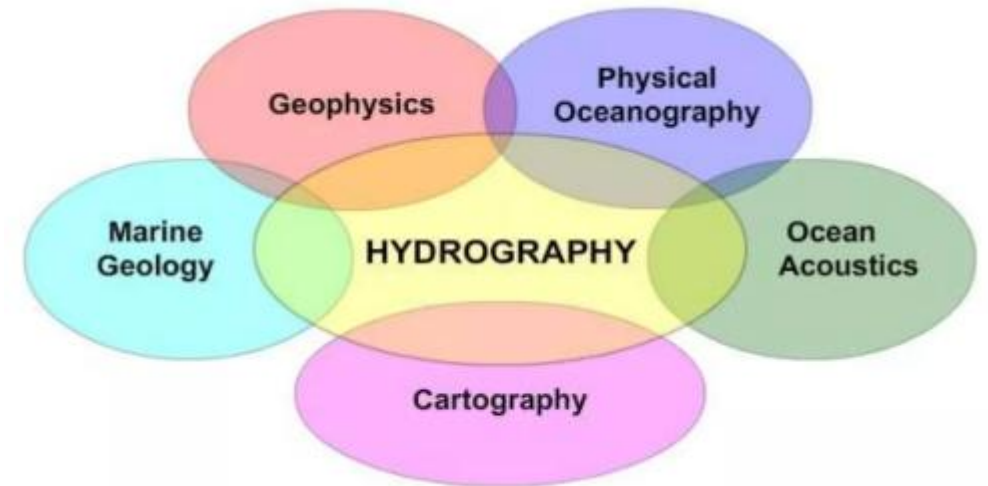
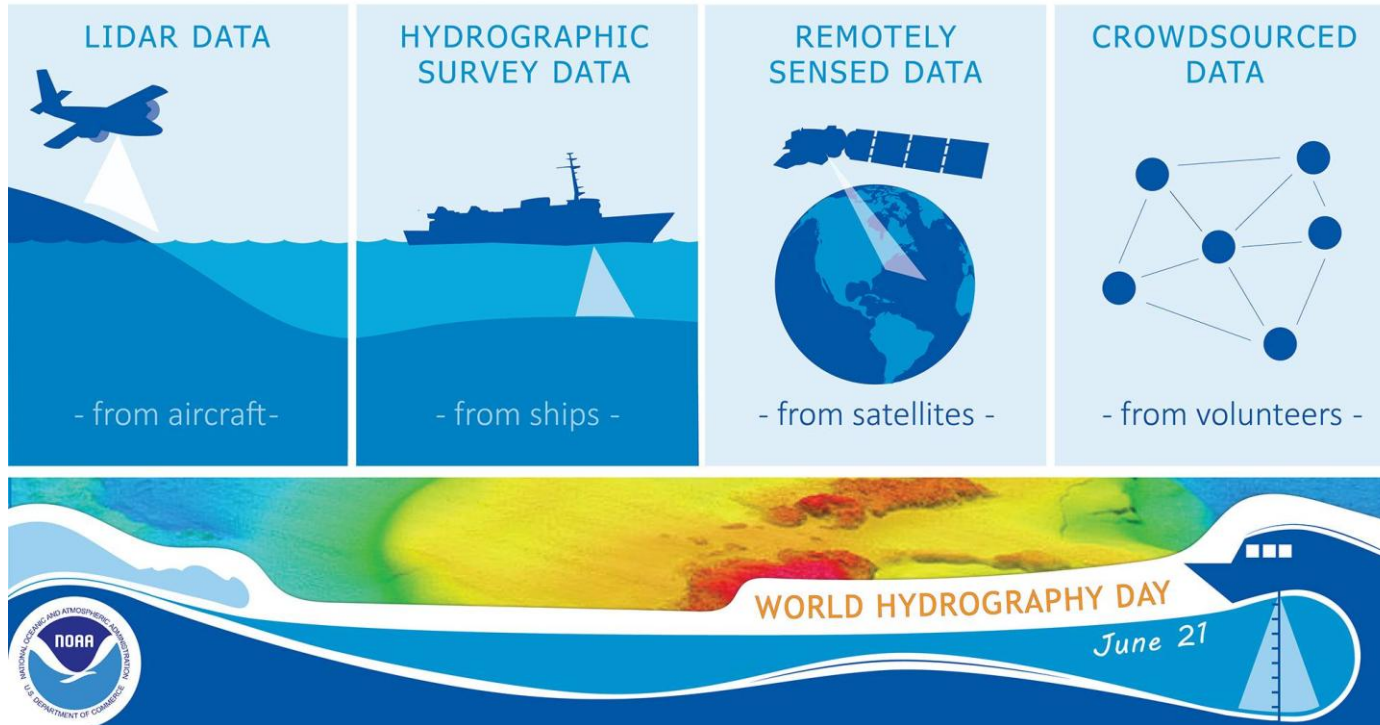
Hydrography includes not only bathymetry, but also the shape and features of the shoreline; the characteristics of tides, currents, and waves; and the physical and chemical properties of the water itself.



CREATING A PICTURE OF THE SEAFLOOR

Bathymetric Data Sources

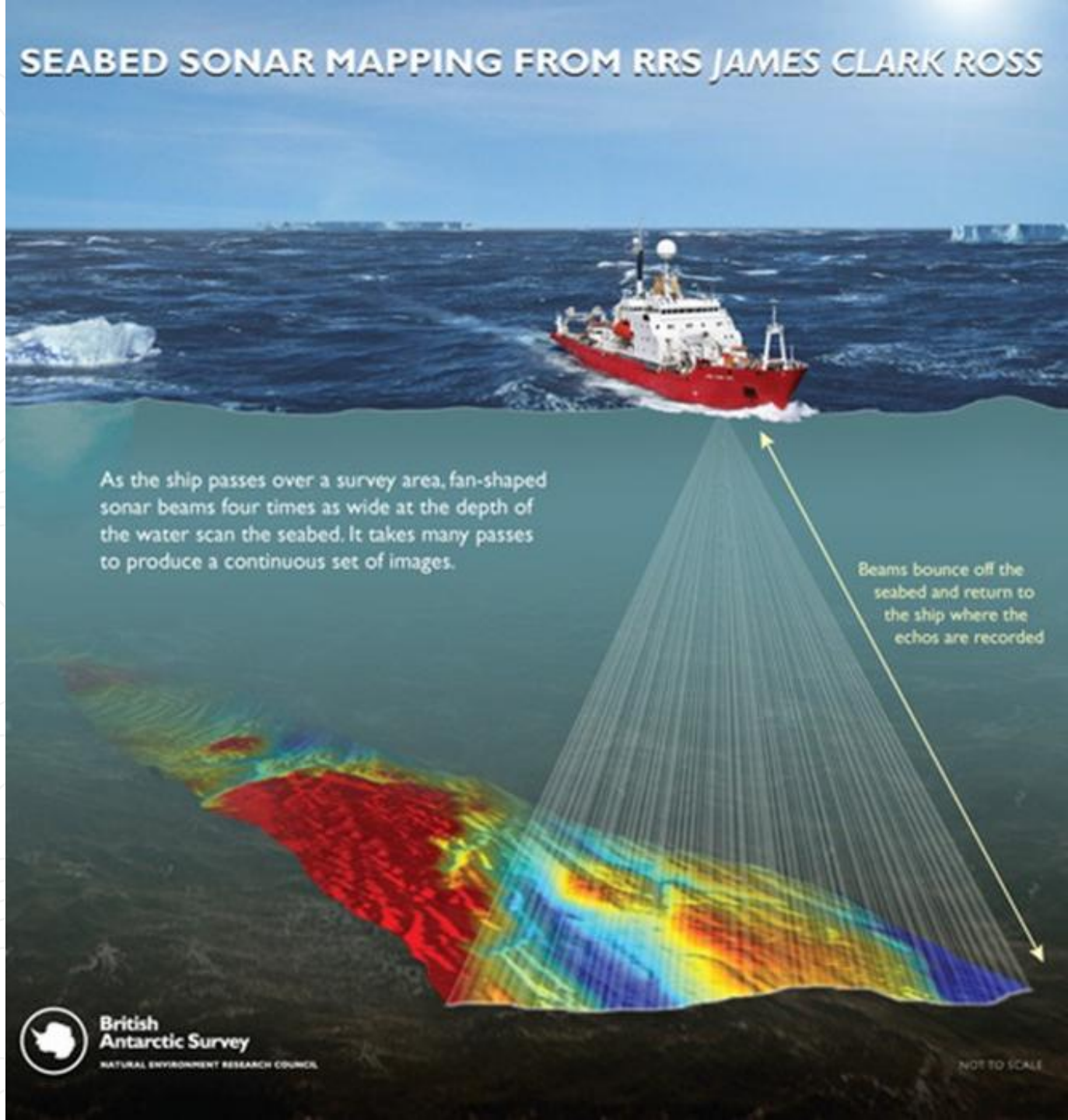
NOAA uses bathymetric data collected from many sources to obtain accurate depths of the seafloor and to update nautical charts.



Source: NOAA

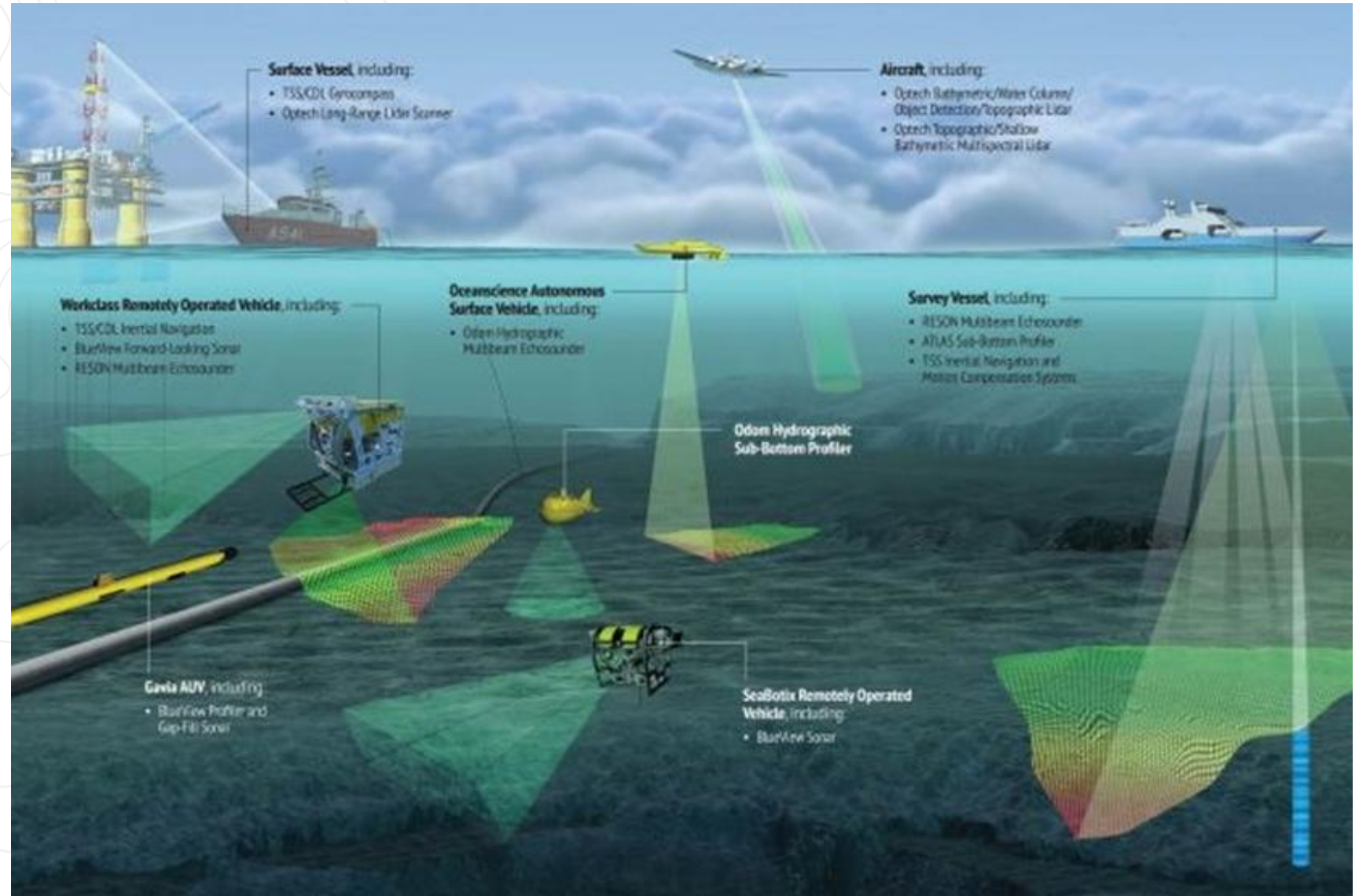
KEY TOOLS

- Inertial Navigation System
- Single Beam echosounder
- Multibeam Echosounder
- Backscatter
- Side Scan Sonar
- Magnetometer
- Sub-bottom profiler
- Bathy Lidar



PLATFORMS

- Survey vessel
- Autonomous Surface Vessel (ASV)
- Autonomous Underwater Vessel (AUV)
- Remotely Operated Vehicle (ROV)
- Aircraft



WHERE WE SURVEY:

Ponds

Rivers

Lakes

Dams

Inlets

Intertidal areas

Nearshore

Offshore



INDUSTRY APPLICATIONS

NAVIGATION AND CHARTING

Ensure safe passage for vessels by mapping underwater terrain and hazards. Supports federal navigation channel surveys and pre/post dredge assessments. NOAA, USACE.

STUDYING CHANGING COASTLINE FEATURES

Monitoring the effects of beach erosion, sea level rise, subsidence, sediment transportation, and mobile underwater hazards.

HYDRODYNAMIC MODELS

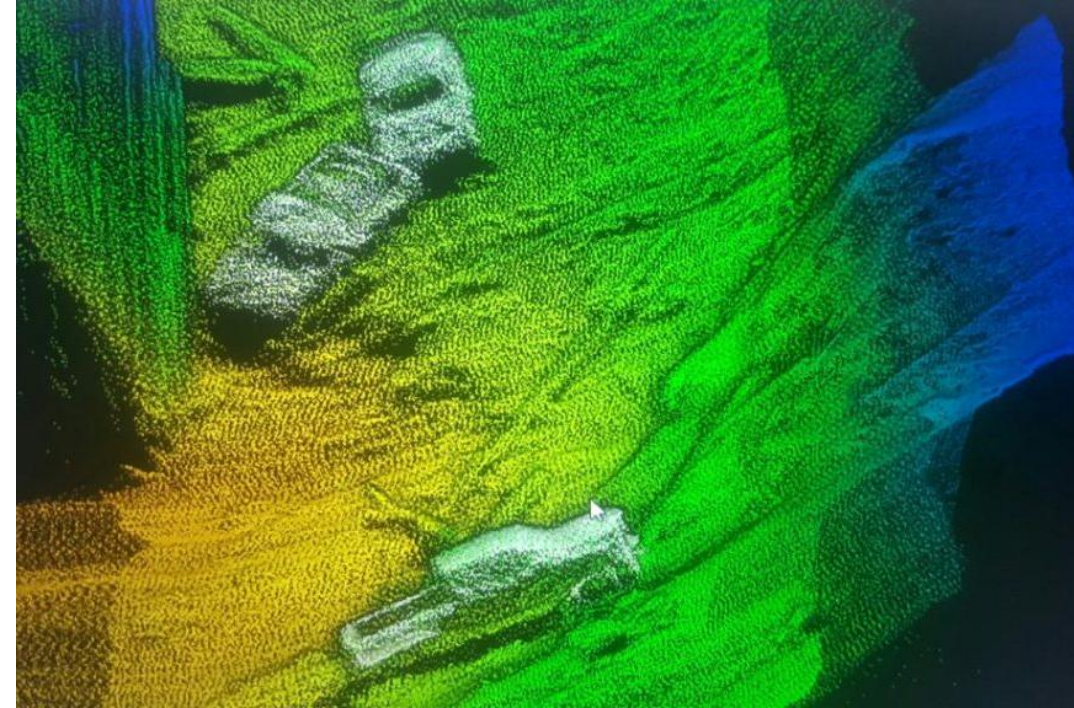
Create models to calculate currents, tides, water temperature, and salinity in an area. These models can also be used to predict tides and currents, as well as hazards like coastal flooding and rip tides.

MARINE CONSTRUCTION & INFRASTRUCTURE

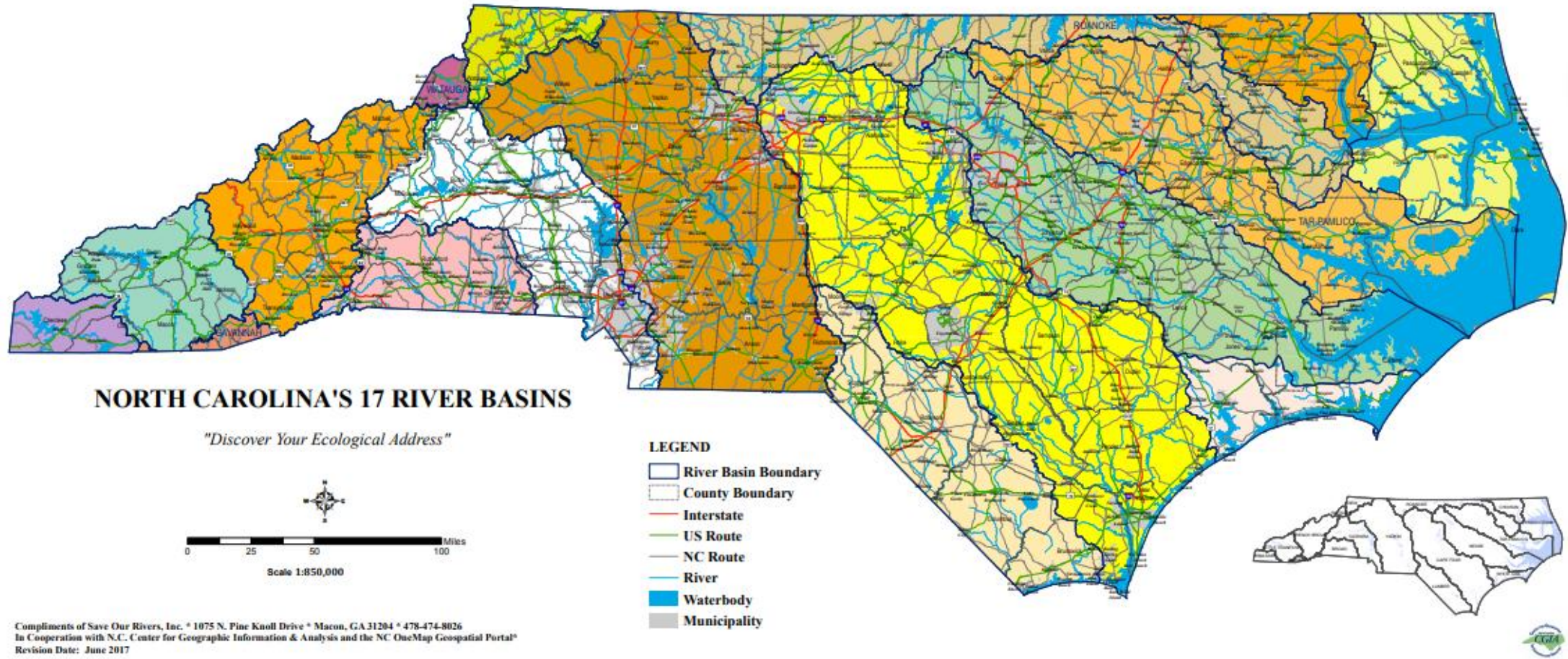
Used to support engineering design, inspection, and remediation for ports, harbors, waterfront development, and other marine infrastructure. Construction as-built (sheet piles, retaining walls and ponds).

STUDYING MARINE LIFE

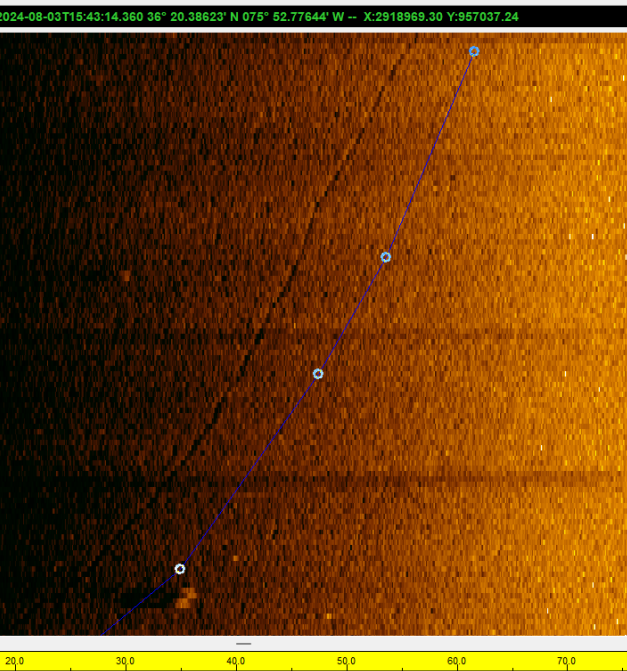
Study the habitats of benthic organisms, to help determine where fish and other marine life feed, live, and breed. Aid in generation of sensitive or protected habitats to assist in conversation and monitoring.



HYDROGRAPHIC SURVEY IN NORTH CAROLINA



<https://www.nconemap.gov/documents/nconemap::river-basins-reference-wall-map/explore?path=>



HYDROGRAPHIC SURVEY APPLICATION EXAMPLES IN NORTH CAROLINA



BATHYMETRIC SURVEY APPLICATIONS ASSOCIATED WITH BOUNDARY SURVEYS

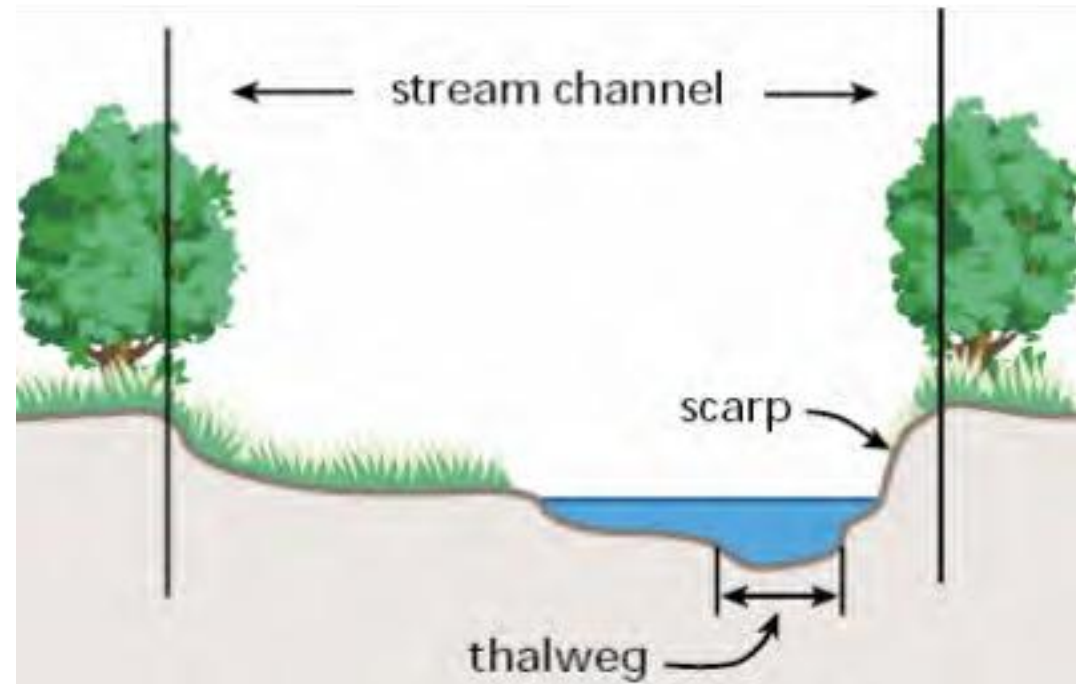
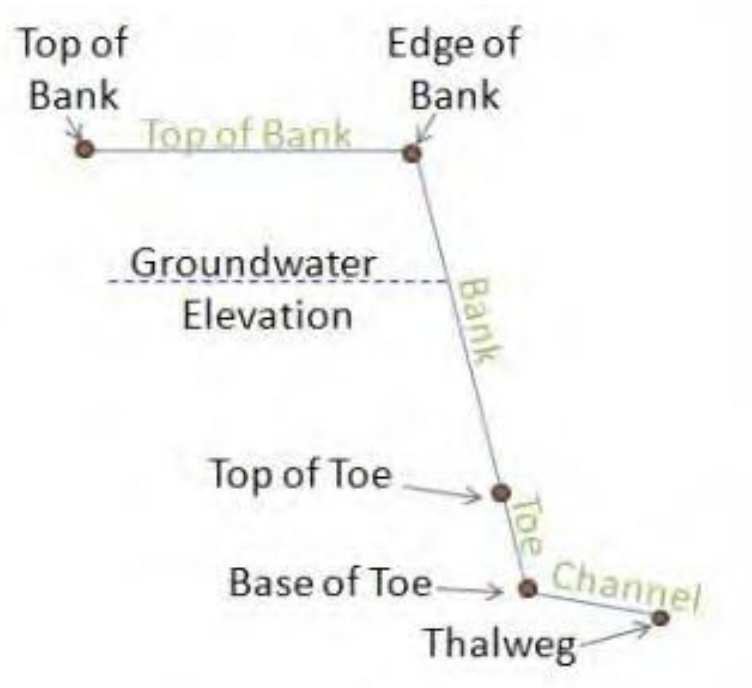
Riparian boundary locations – high water mark.

Accretion and erosion.

Support permitting for structures near/on water (piers, docks, living shorelines, underwater utilities).

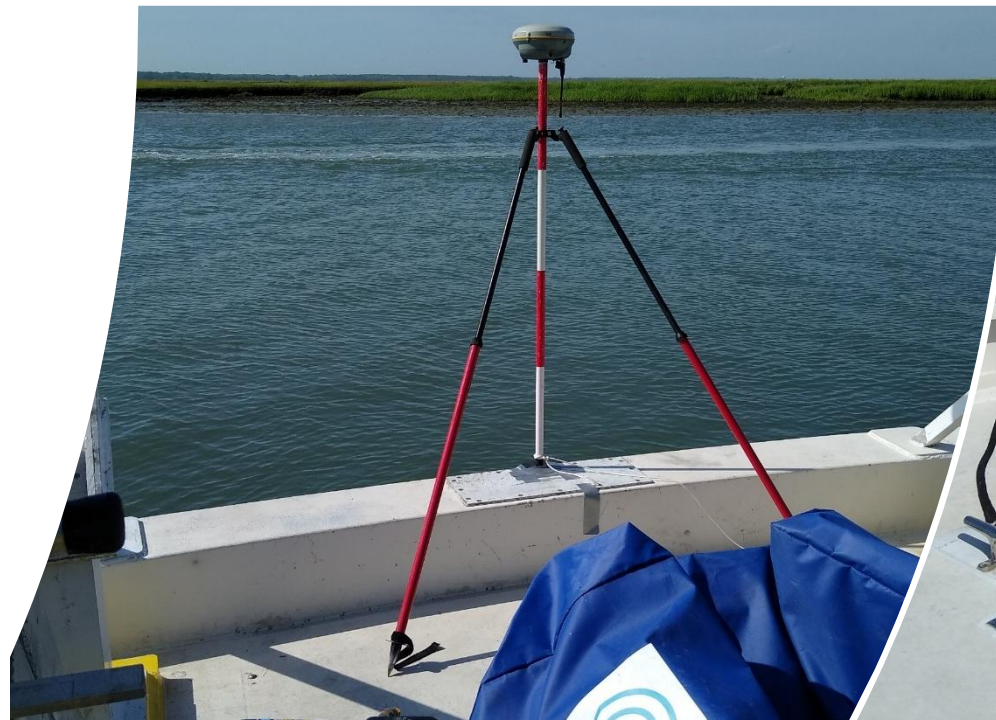
Identify lease areas.

Determine channel thalweg.

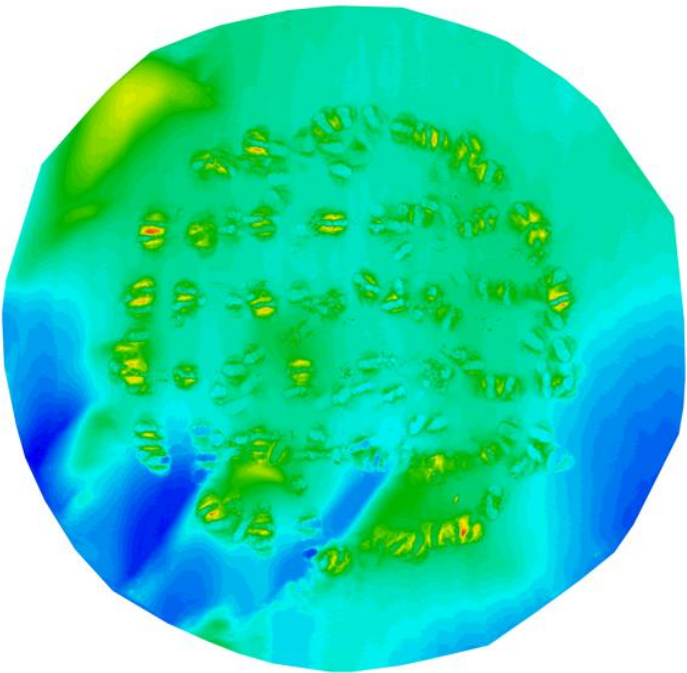
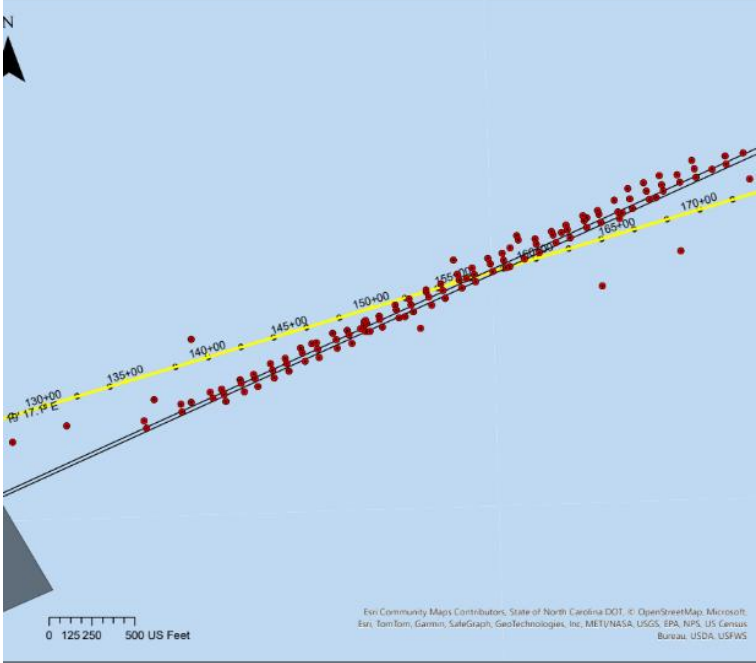
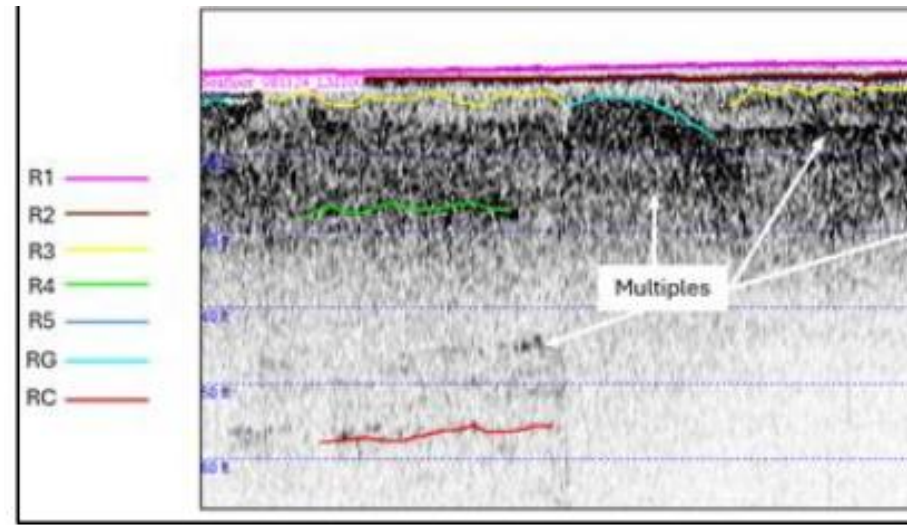
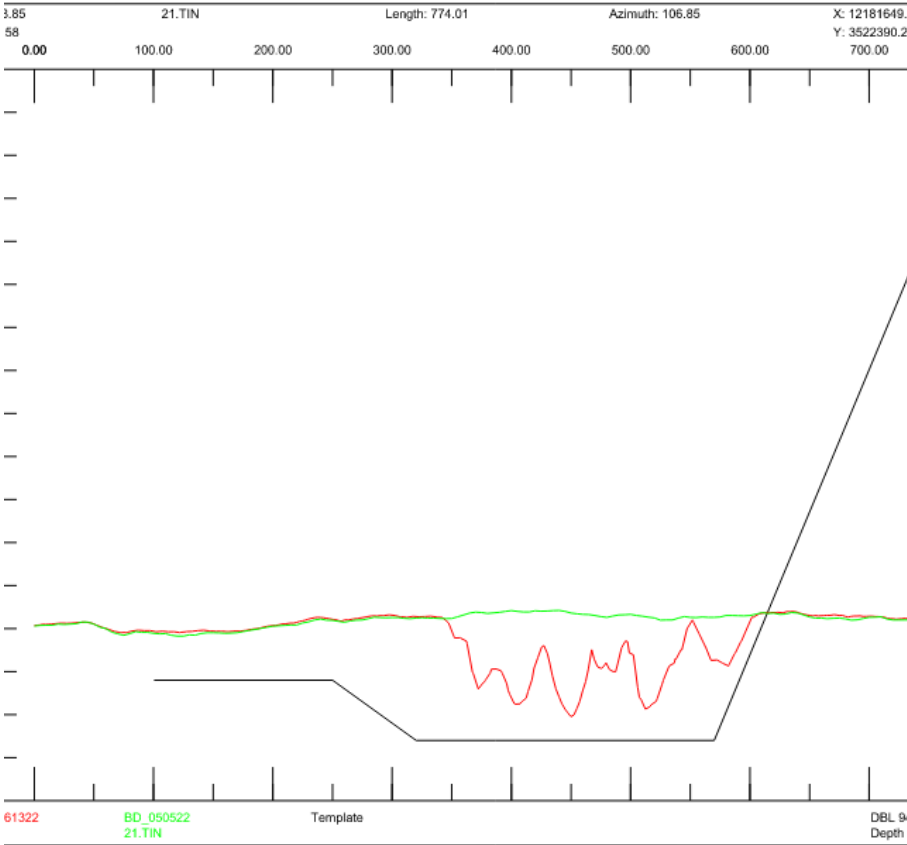


LAND SURVEY EQUIPMENT AND PRACTICES IN BATHYMETRIC SURVEY

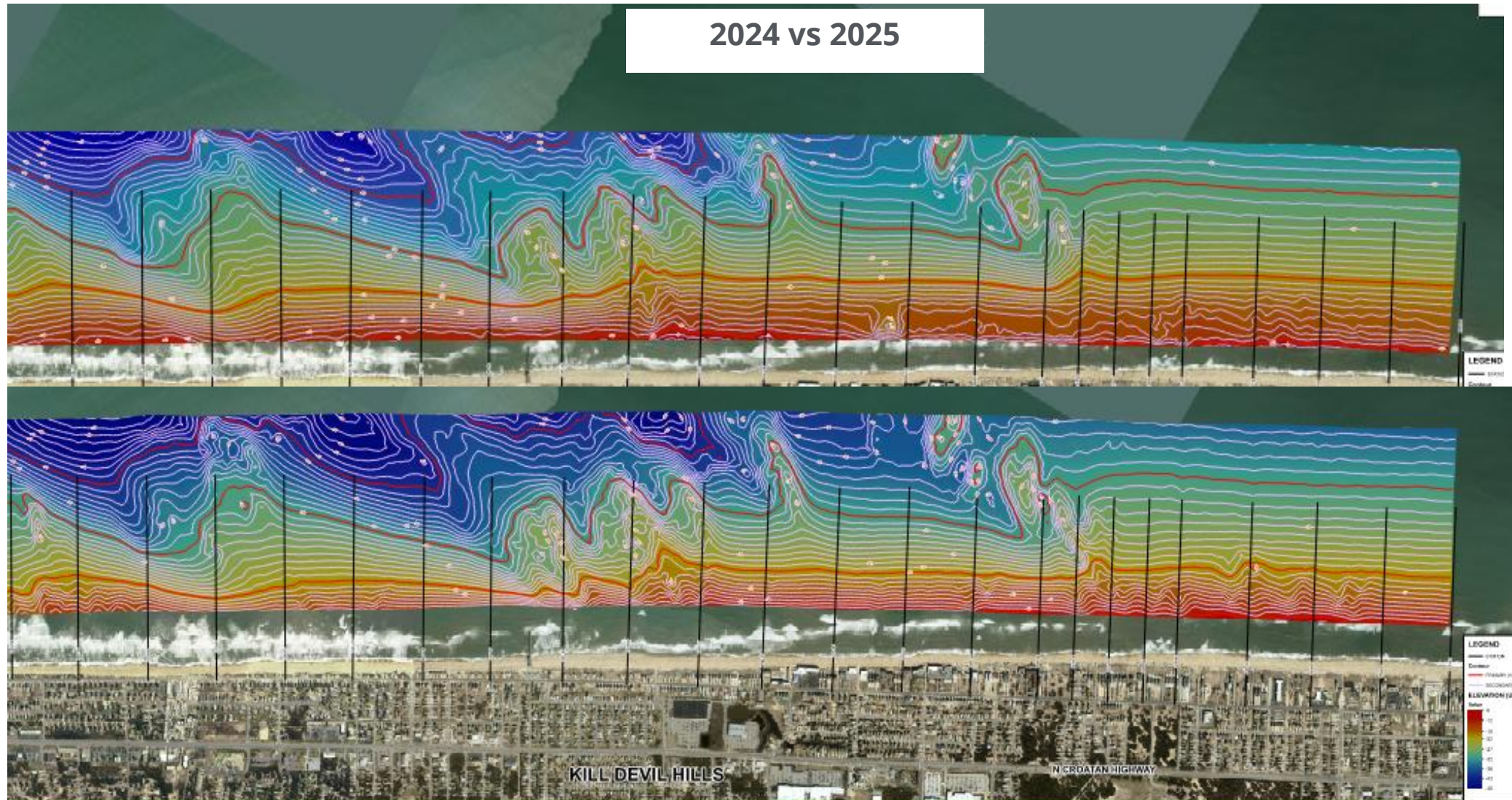
- Water level monitoring
- Dimensional Control Surveys
- 3D position checks
- Heading and alignment checks
- Independent “soundings” by poling or other physical measurement methods
- Overlap transition zones from topographic, aerial, or lidar data sets



DREDGING AND MARINE CONSTRUCTION WORKS APPLICATION EXAMPLES



COASTAL ENGINEERING APPLICATION EXAMPLE



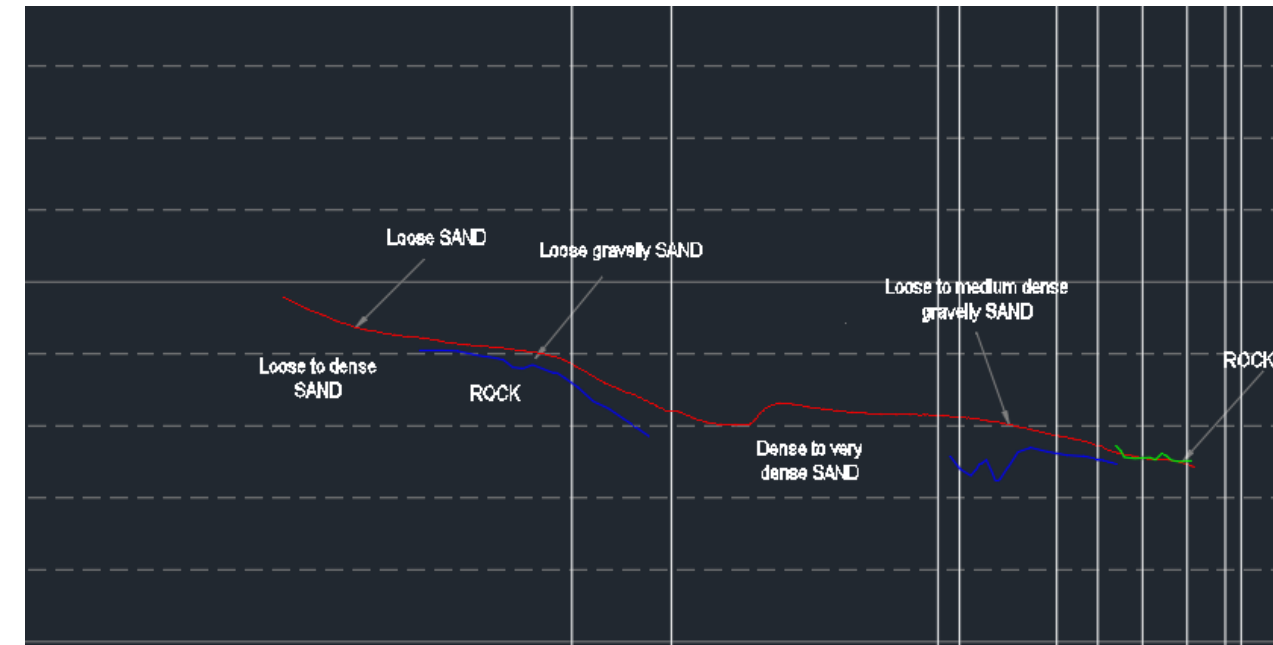
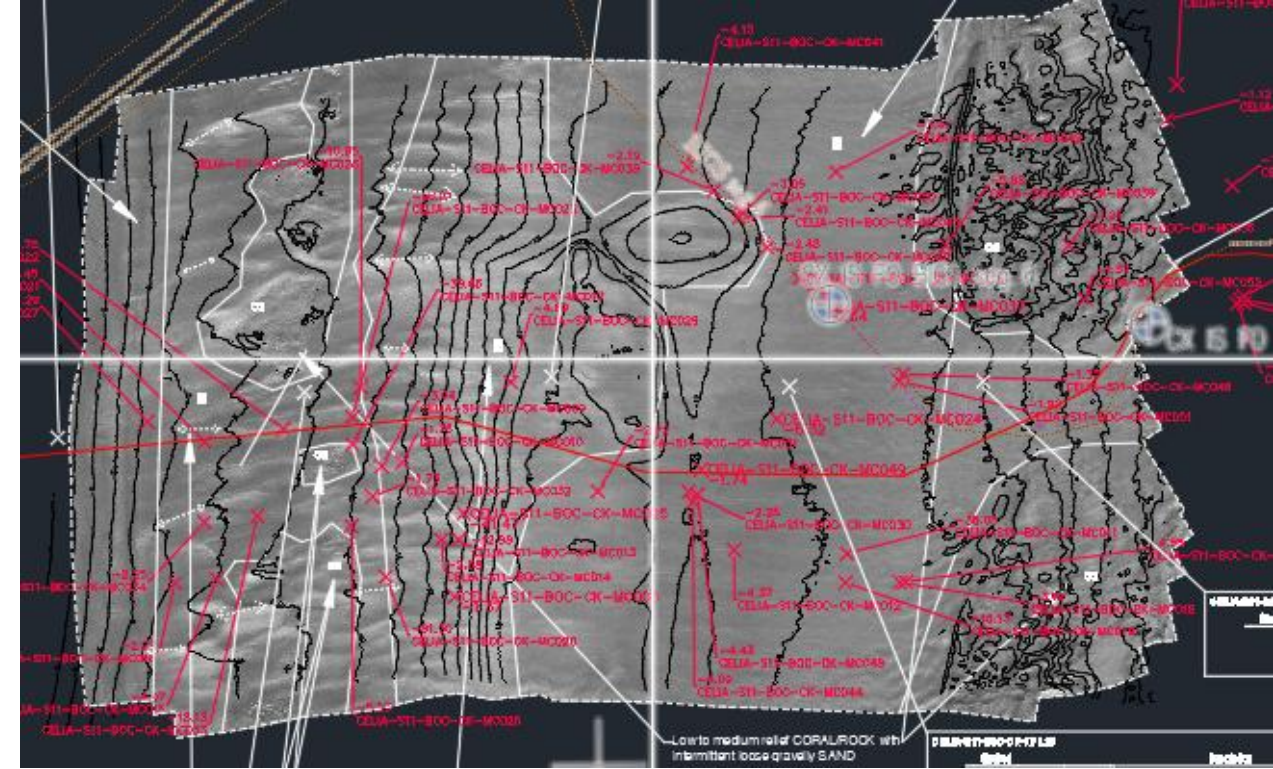
Scour monitoring and sediment transportation studies of critical infrastructure

CABLE LANDING AND CULTURAL RESOURCES SURVEY - STATE HISTORICAL PRESERVATION OFFICE (SHPO) SURVEYS & INVESTIGATIONS

Must be able to:

- Detect and characterize shipwrecks and debris (sonar imagery + magnetics)
- Resolve buried features (CHIRP/sub-bottom profiles)
- Map subsurface stratigraphy
- Provide accurate positioning of anomalies (special-order hydrographic accuracy ≤ 2 m)
- Deliver high-resolution digital mosaics, interpreted charts, and anomaly tables

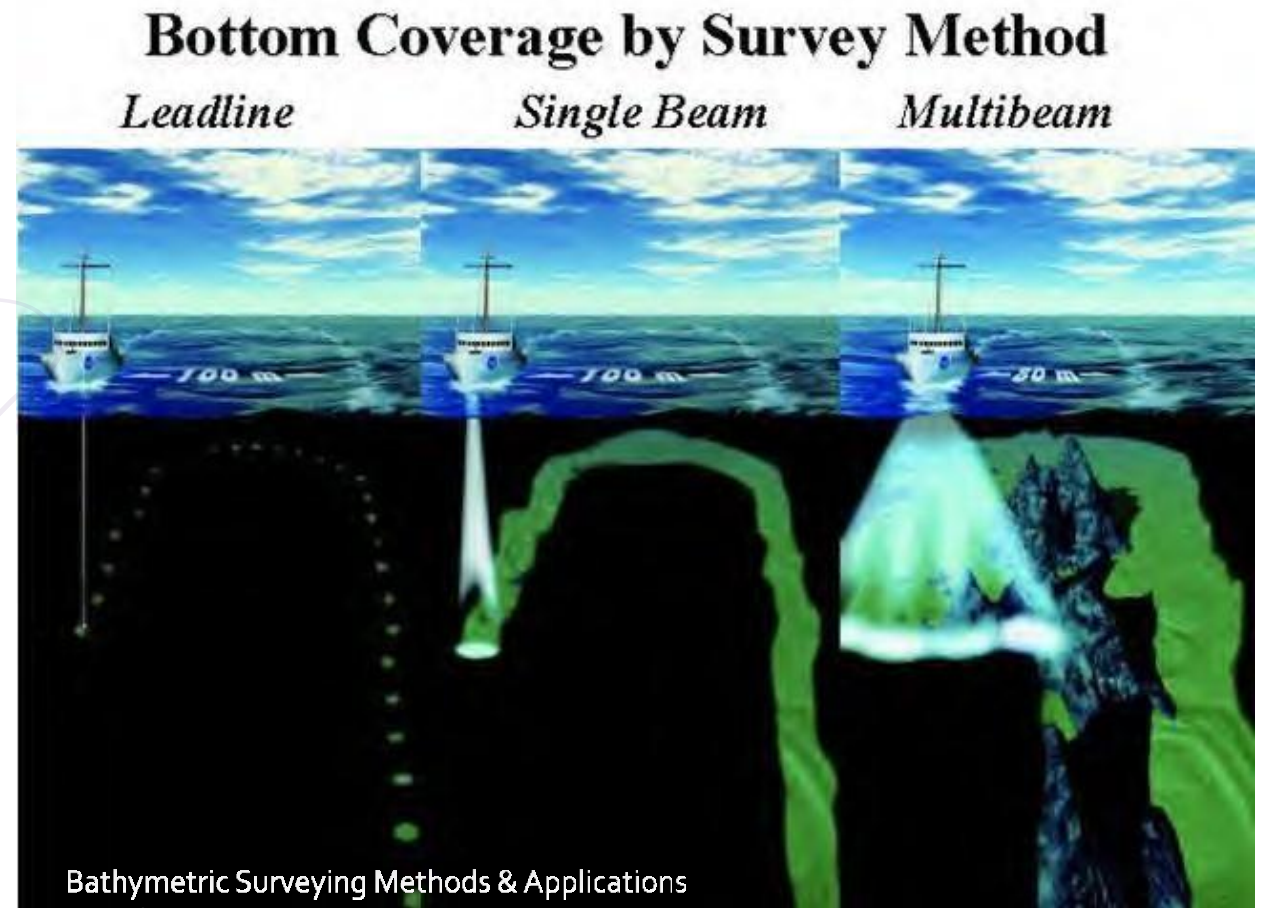
Partnership with Archaeologists, geologists and geophysicists for data collection and analysis



BATHYMETRIC SURVEY METHODS

Similar to terrestrial and aerial surveying, achieving higher resolution, broader coverage, and faster mapping comes with increased equipment costs.

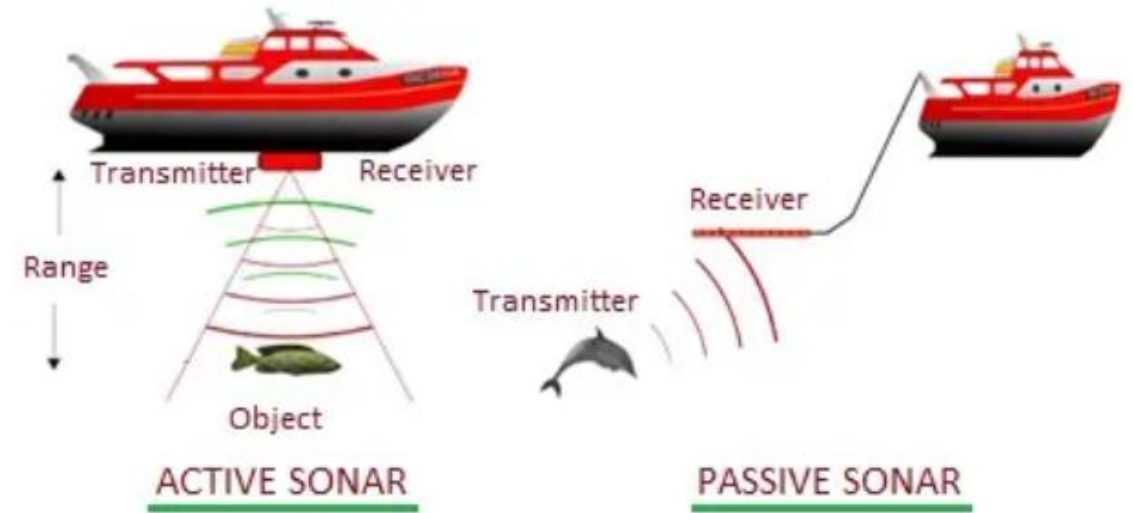
- Lead line, pole surveys, or “topo shots” = \$
- Single-beam sonar surveying = \$\$
- Multibeam echosounder surveying = \$\$\$



KEY DEFINITIONS:

Sonar (**SO**und **N**avigation **A**nd **R**anging)

A method or device for detecting and locating objects underwater by means of sound waves sent out to be reflected by the objects. It measures the time it takes from when the ping leaves the sonar until it returns.



Sonar Components:

TRANSDUCER

Converts electrical energy into sound pressure waves or vice versa

HYDROPHONES/RECEIVERS

Captures and amplifies the echoes reflected from objects in the water

PULSE GENERATOR/PROJECTORS

Generated pulses of sound waves sent into the water

DISPLAY AND ANALYSIS SYSTEM

Processes the received signals to determine the distance, direction, and characteristics of underwater objects.

SINGLE BEAM DEPTH MEASUREMENT SYSTEMS

REPLACED MANUAL DEPTH MEASUREMENTS IN THE 1960'S

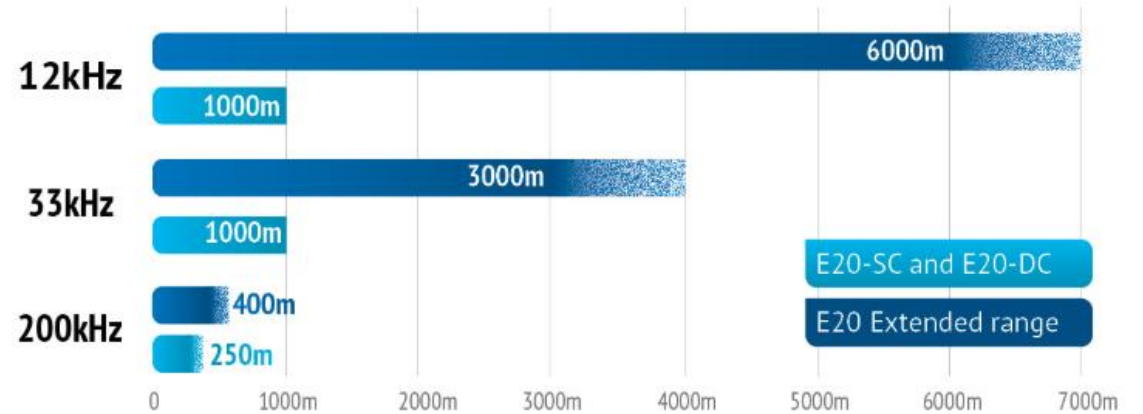
SYSTEMS

Single frequency
Dual frequency
Multiple transducer channel sweep systems

APPLICATIONS

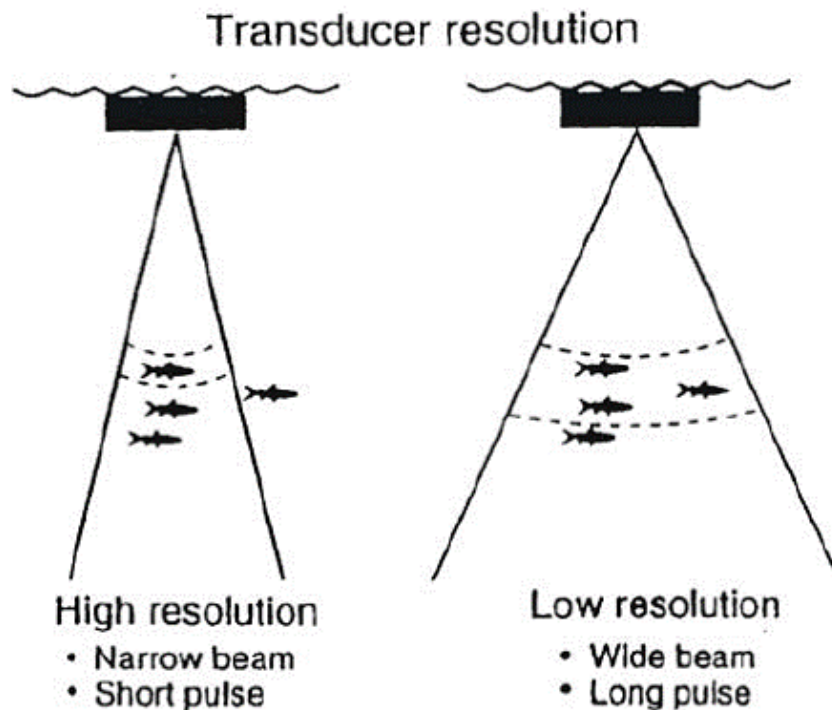
Shallow draft navigation projects
Beach monitoring/nourishment projects
General condition surveys

Teledyne Odom E20 depth range



SINGLE BEAM DEPTH MEASUREMENT SYSTEMS

Wider beam transducers are smaller and less affected by the vessel motion



Attribute	Wide beam	Narrow beam
Depth penetration	lower	higher
Horizontal extent	higher	lower
Horizontal resolution at depth	lower	higher
Near-field	lower	higher
Deadzone	higher	lower
Ambient noise levels	higher	lower

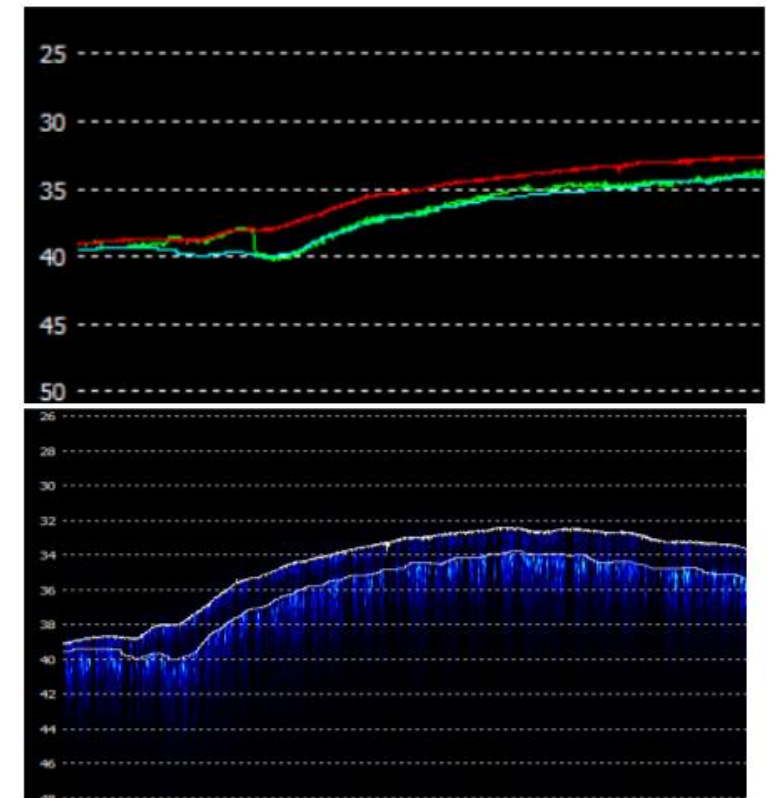
SINGLE BEAM DEPTH MEASUREMENT SYSTEMS

OPERATION SETTINGS

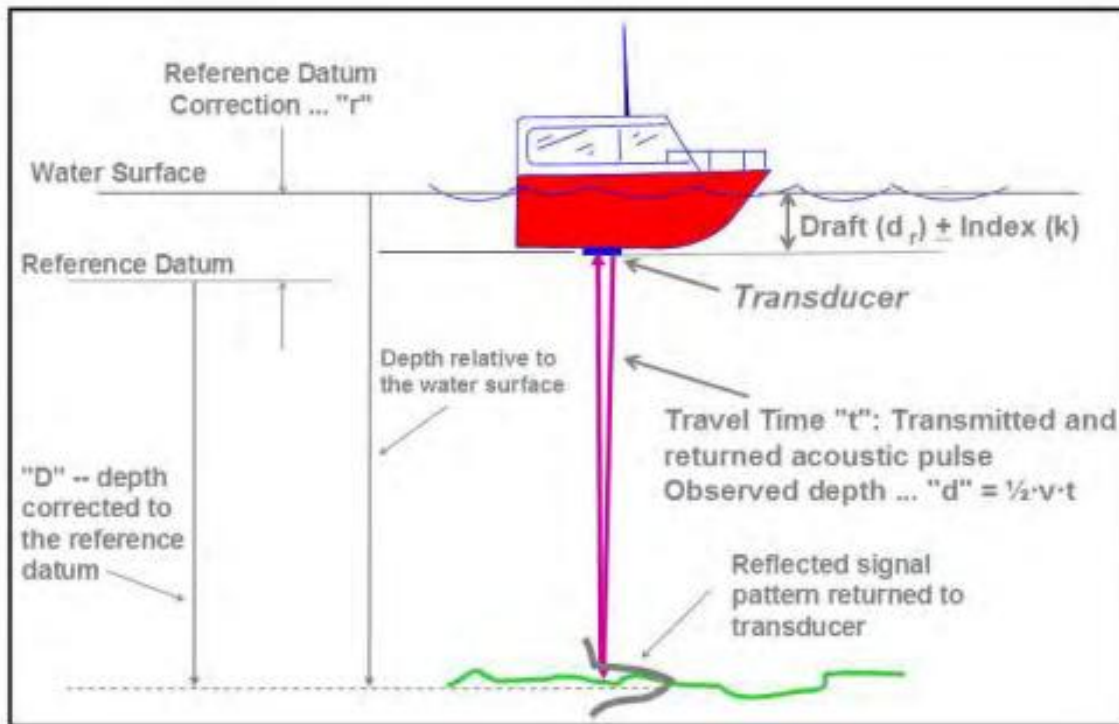


- ❖ **Power:** Should be kept at the lowest consummate with adequate detection (*depth, sediment*)
- ❖ **Gain:** Amount of amplification/attenuation applied to the return echo signal.
- ❖ **Pulse:** Determines the vertical resolution of the echosounder – **short pulses=higher resolution** (*ability to separate the return from different but nearby objects*)
- ❖ **Range:** Where to look for the bottom
- ❖ **Max Ping Rate:** Number of sounding per second

HIGH VERSUS LOW FREQUENCY



SINGLE BEAM DEPTH MEASUREMENT SYSTEMS



Depth corrected to reference datum: $D = \frac{1}{2} (v \cdot t) + k + d_r + r$ (Eq 4-1)

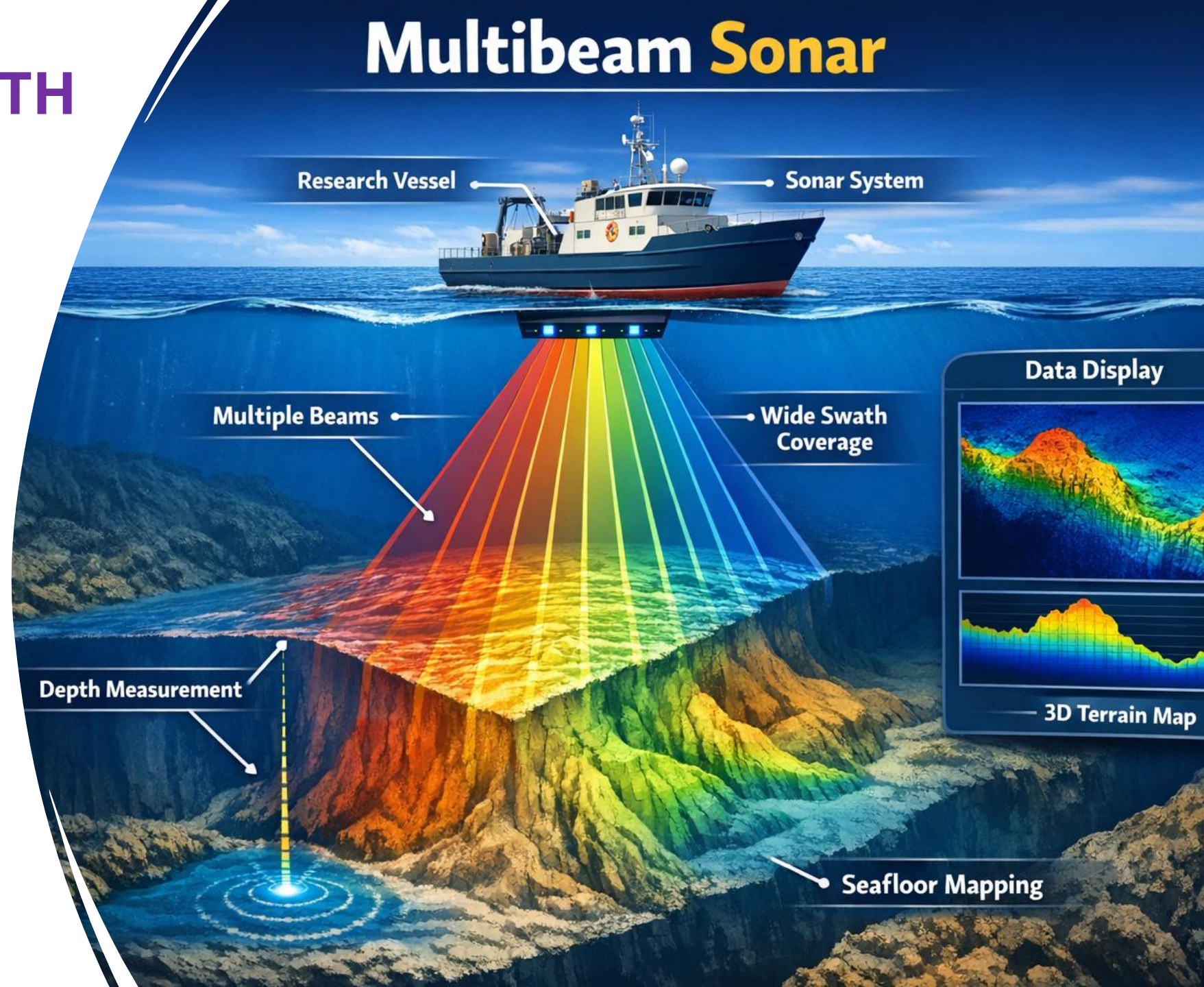
where:

- v = average velocity of sound in the water column
- t = measured elapsed travel time from transducer to bottom and back to transducer
- k = system index constant
- d_r = distance from reference water surface to transducer (draft)
- r = reference datum correction ("tide" or "stage" correction)

**This formula does not take the motion into account.*

MULTI BEAM DEPTH MEASUREMENT SYSTEMS

- Developed for shallow water in the 1980's
- The multibeam systems replaced single beam in most deep-water navigation projects
- 100% bottom ensonification coverage
- Typically 170-700 kHz for high resolution mapping



MULTI BEAM DEPTH MEASUREMENT SYSTEMS

PRINCIPLES

Amplitude/Phase detection (MB2)

Interferometric systems (phase comparison) (6205)

Beam Spacing (*equiangular vs equidistant*)

Beam footprint size (*beam width*)

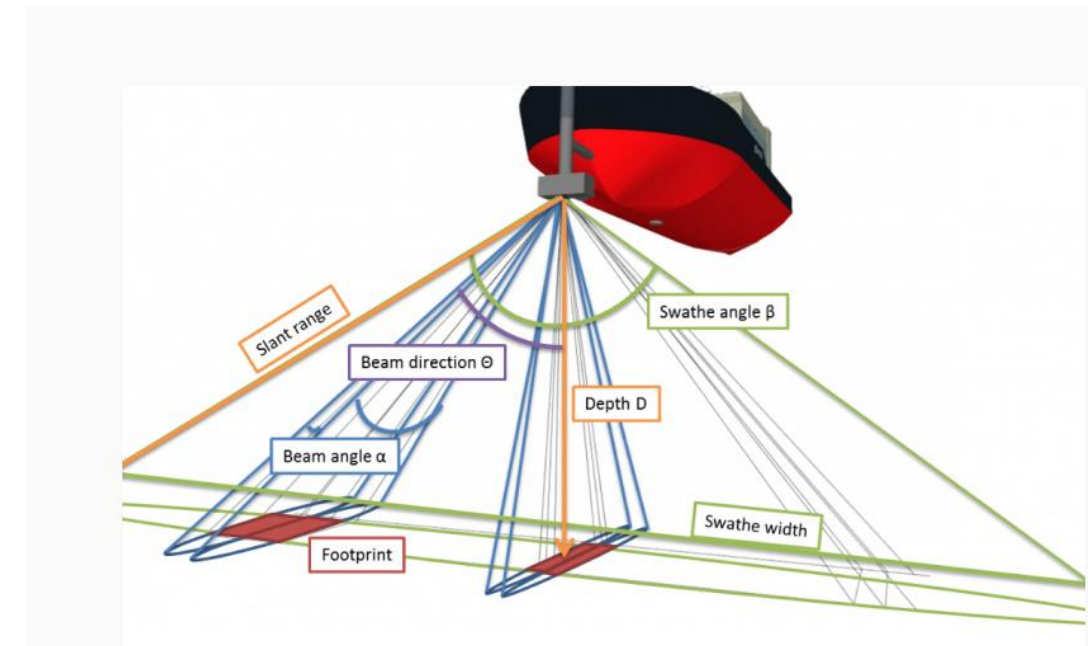
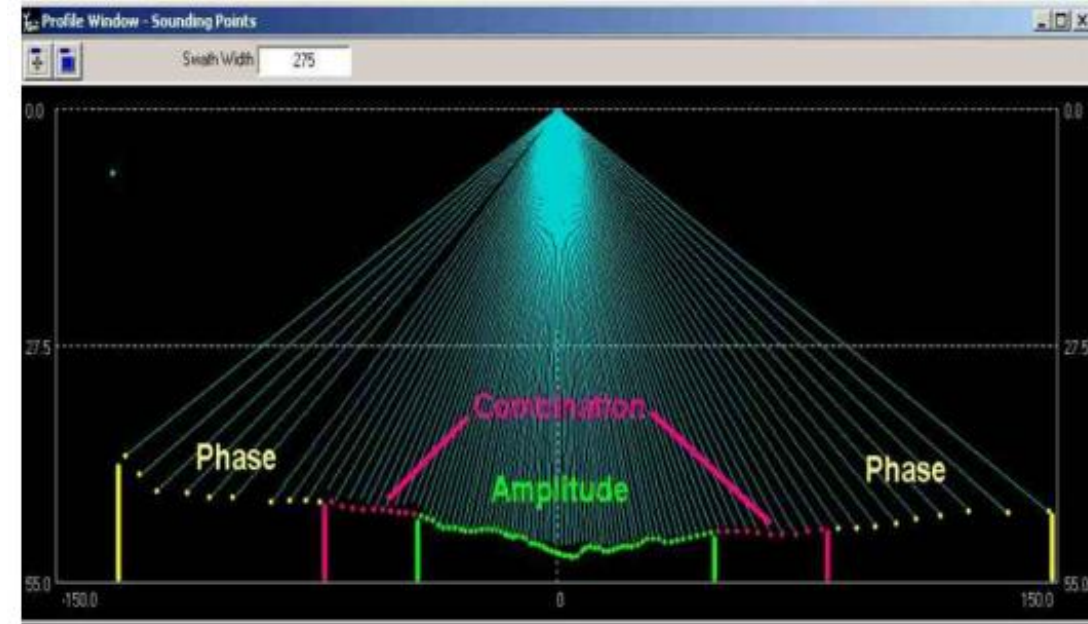
Signal parameters – dependent on bottom reflectivity and slope angle of incidence

Vessel positioning requirements

TRIMBLE RTX

Precise corrections in real-time. Every time.

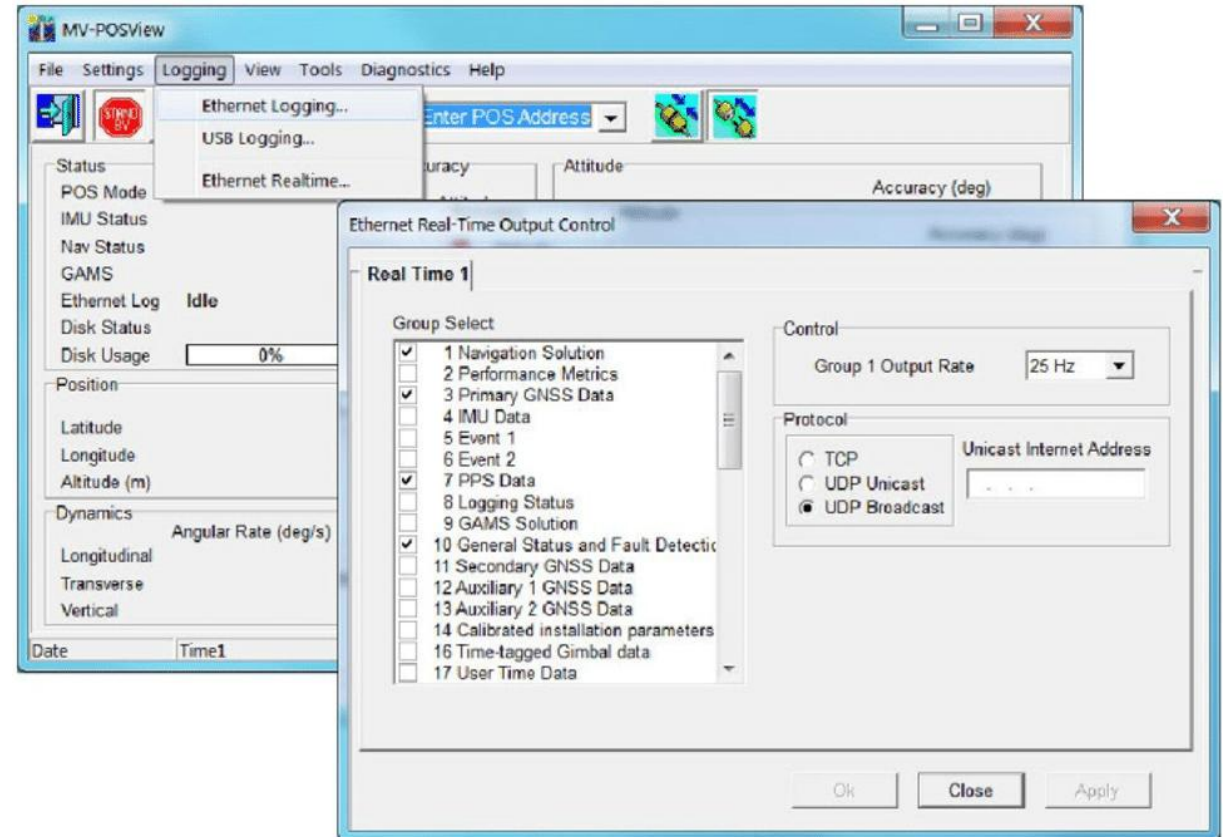
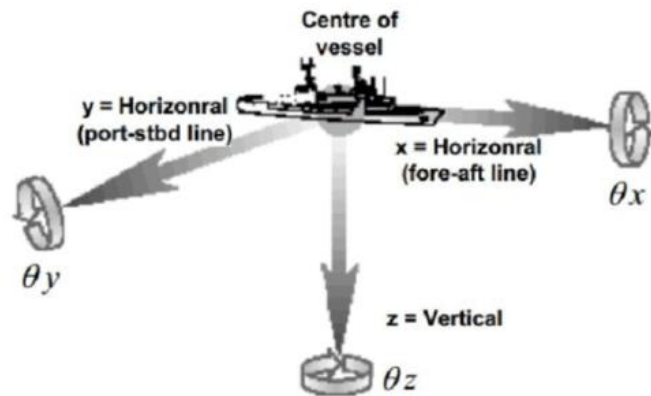
Trimble RTX[®] is our exclusive, advanced precise point positioning technology that provides real-time, centimeter-level corrections. Experience true mobility working without the constraints of a local base station or VRS network, with high-level accuracy delivered to you worldwide via satellite or cellular/IP.



MULTI BEAM DEPTH MEASUREMENT SYSTEMS

MOTION

Correcting for motion ([heave, roll, pitch and yaw](#))
– Inertial navigation systems



MULTI BEAM DEPTH MEASUREMENT SYSTEMS

SOUND VELOCITY AND REFRACTION

- **Depth conversion:** Converting acoustic travel times to depth using the harmonic mean, not average or in situ sound speed.
- **Ray tracing:** Determining the true path of sound waves in non-vertically incident beams.
- **Beam steering:** Ensuring accurate steering angles at the transducer face in multibeam echosounding

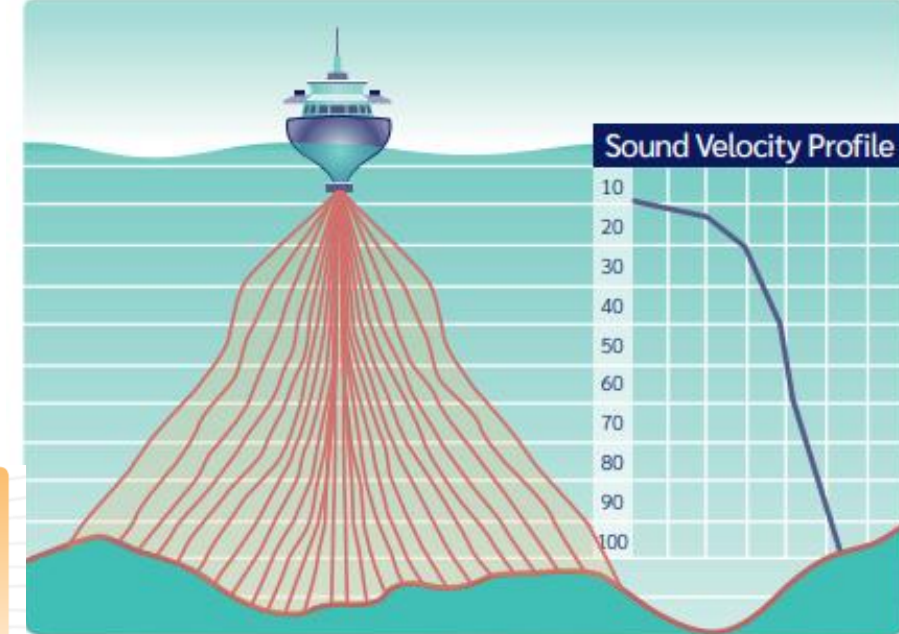
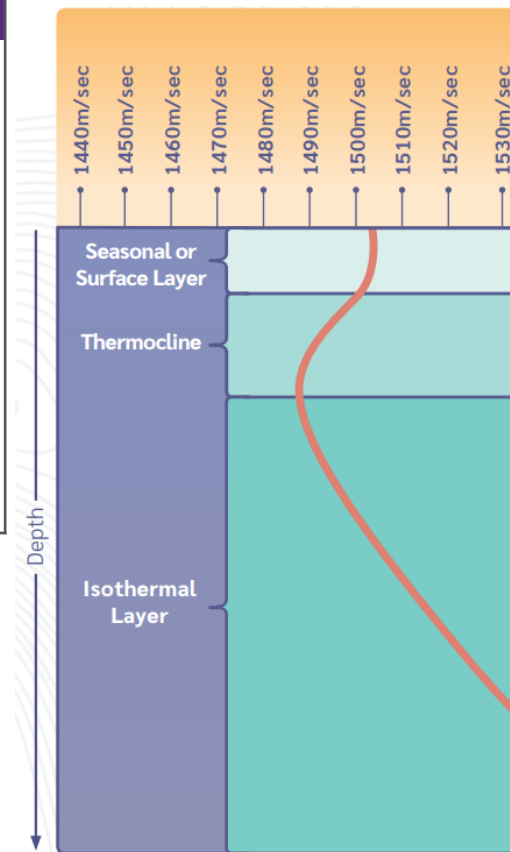
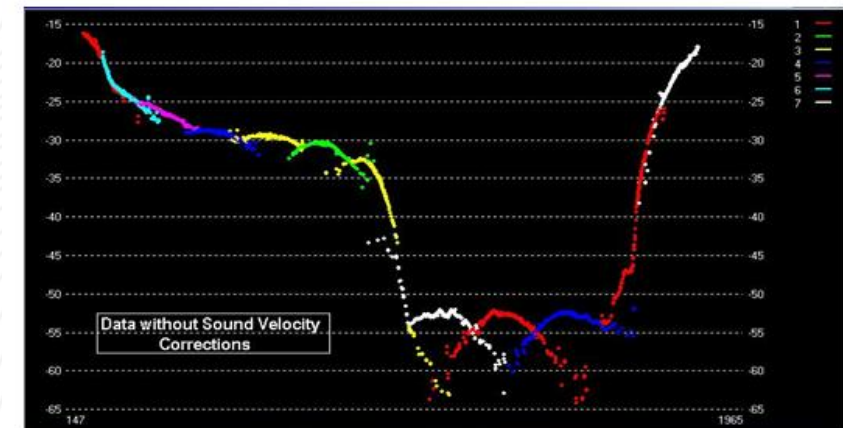


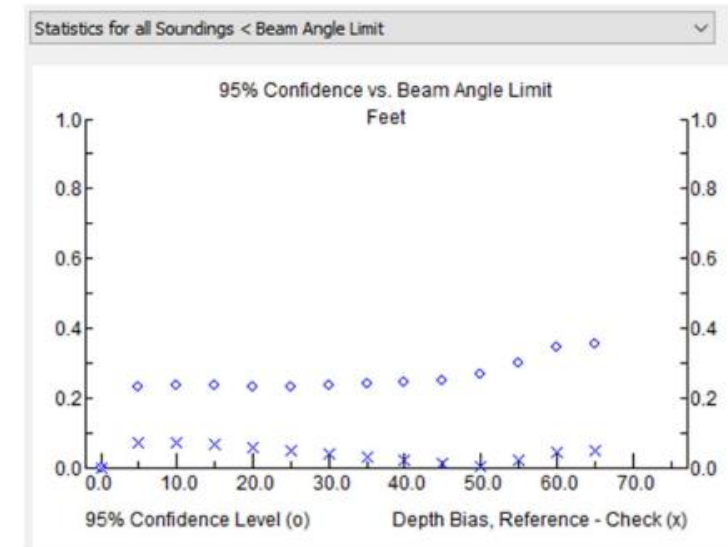
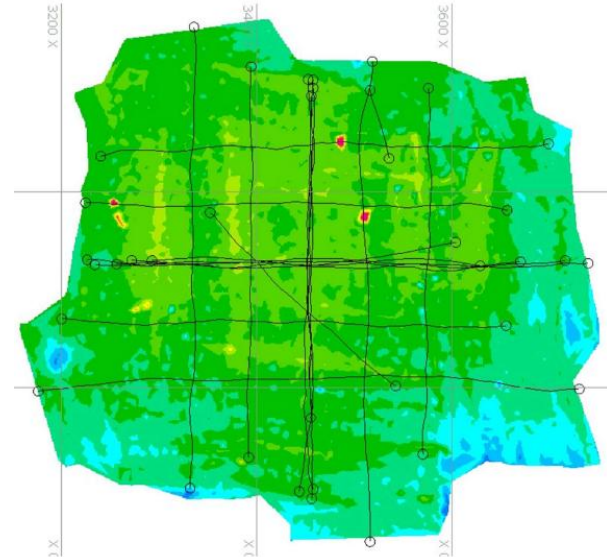
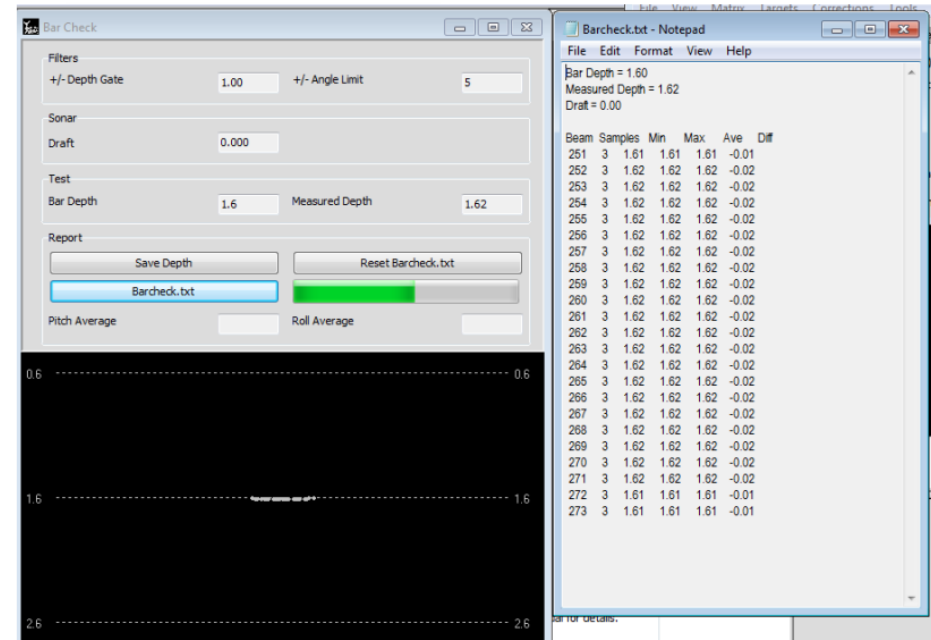
Figure 2: Concept of refraction due to different sound velocities in the water column

1 °C change in temperature	=	4.0 m/sec change in velocity
1 ppt change in salinity	=	1.4 m/sec change in velocity
100 m change in depth (10 atm's pressure)	=	1.7 m/sec change in velocity



SYSTEMS CHECKS AND CALIBRATION PROCEDURES

- Control position vs GNSS Rover
- Water surface elevation with GNSS Rover vs Hydrographic system
- Bar check
- Cross-check lines: single beam and outerbeams
- Performance test: outerbeams
- GPS Health Check
- Gyro Calibration



95% Confidence vs. Beam Angle Limit in Feet

SYSTEMS CHECKS AND CALIBRATION PROCEDURES

➤ Patch test

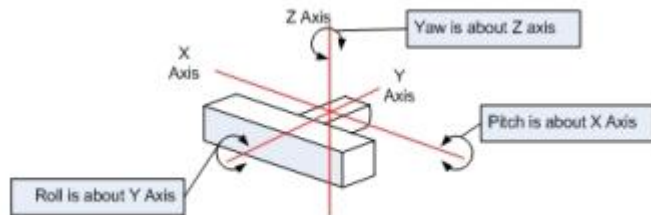


Figure 1: Sonic 2024/2022 axes of rotation

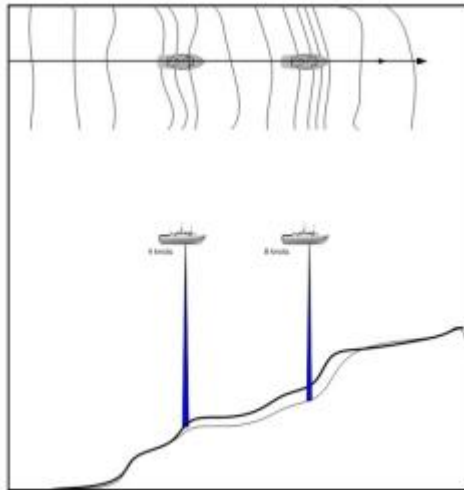


Figure 2: Latency Data collection

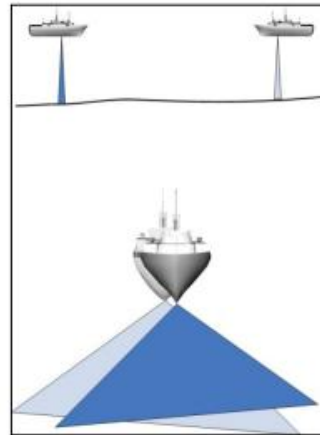


Figure 3: Roll data collection



Figure 4: Roll data collections

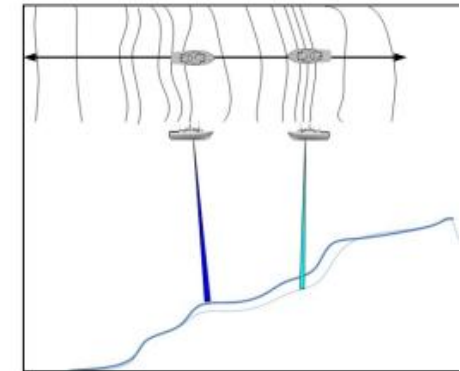


Figure 5: Pitch data collections

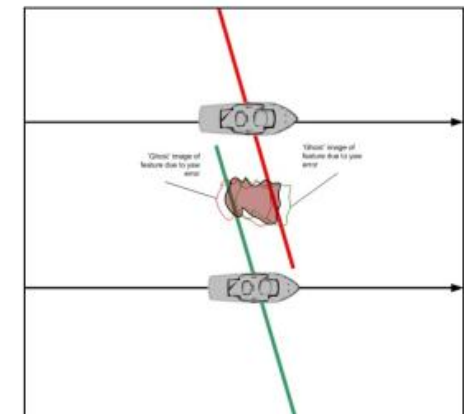
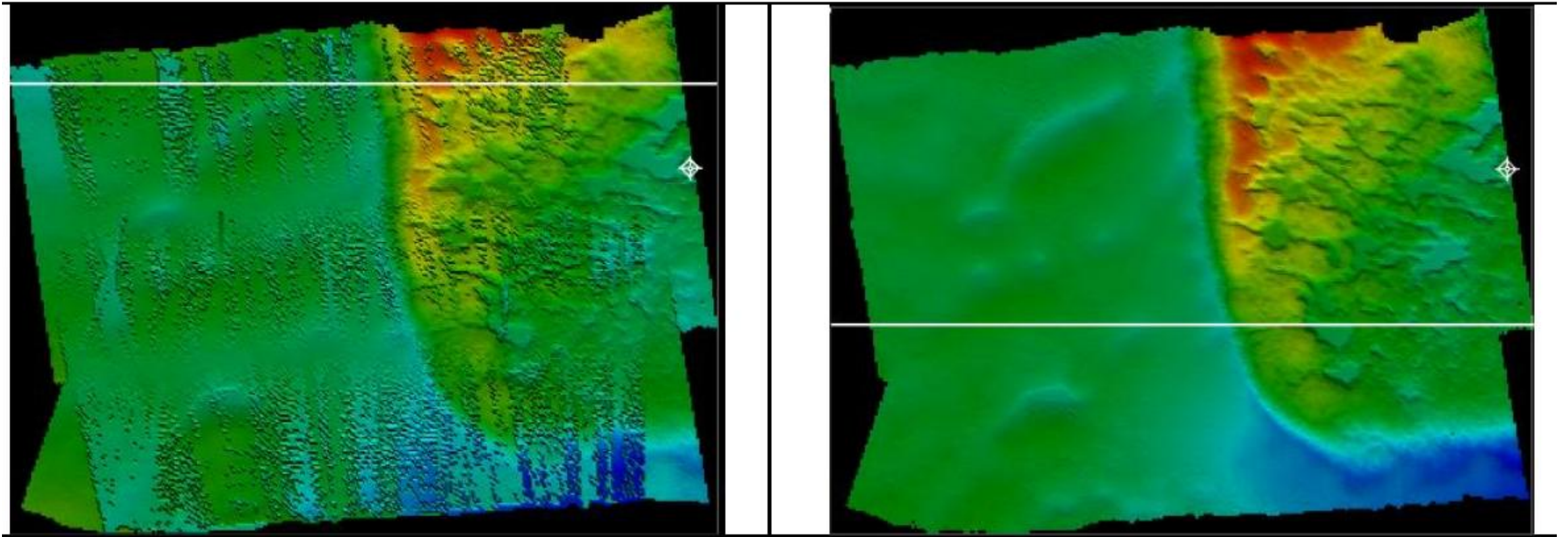


Figure 6: Yaw data collection

SYSTEMS CHECKS AND CALIBRATION PROCEDURES

➤ Patch test



Before vs After

BATHYMETRIC SOFTWARE

ACQUISITION SOFTWARE AND NAVIGATION PACKAGES ARE NECESSARY TO:

- Record the data
- Provide real time QA/QC
- Assist with infill and planning

COMMONLY USED HYDROGRAPHIC SURVEY SOFTWARE:

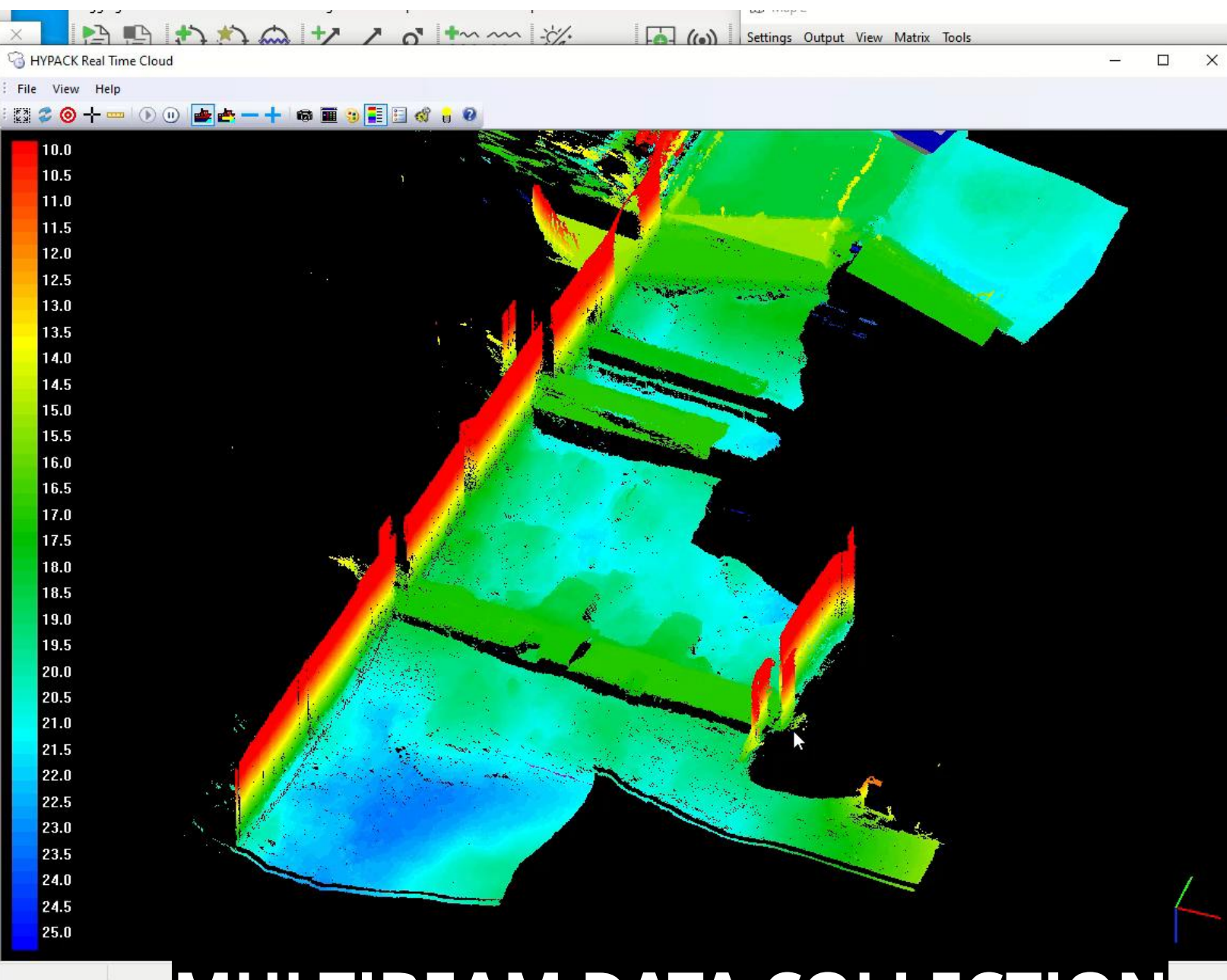
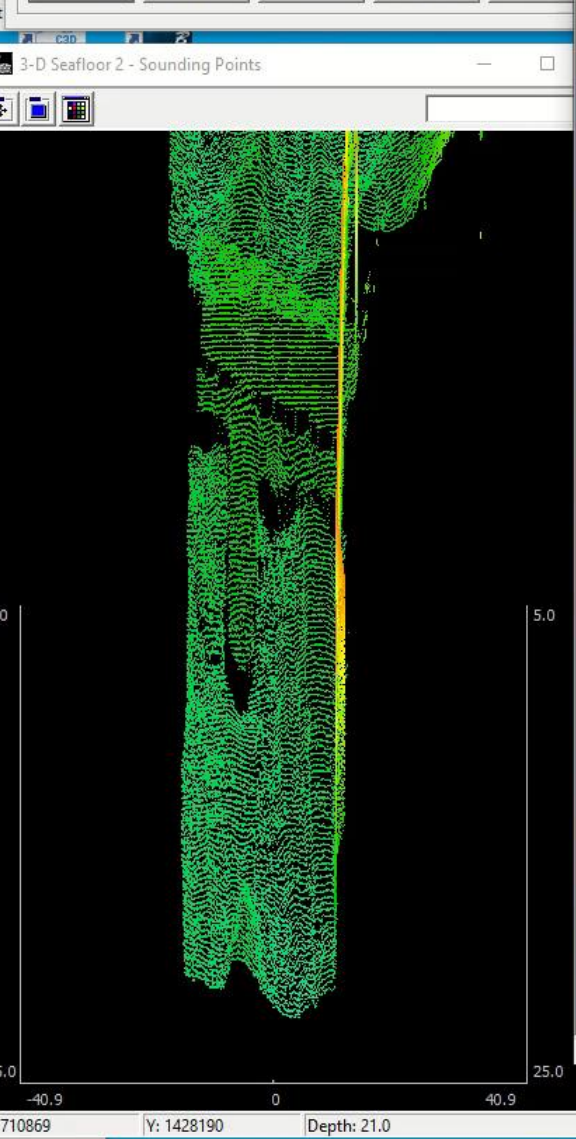
- Hypack / HySweep Survey
- QPS QINSy
- CARIS
- Sonawiz



Playback Controller

File 0002_1127.HSX

Pause Play > Fwd >> Search... End



18.2

73d41'06

NOT LOGGING

13:41:51

1

185.75

19.16°

16.50

-2.12

^0.7

27:49:44

0.00

MULTIBEAM DATA COLLECTION

MULTI BEAM SYSTEM ACCURACY

ESTIMATED ACCURACY

- Depth measurements contain a lot of uncertainties, so measuring and verifying that the survey met a “±” dispersion is problematic.
- We can verify repeatability by means of performance test, cross lines.

**Performance Tests on a dredge clearance survey "should be repeatable to 0.2 ft. and that the 95% standard deviation not exceed (±0.5 ft)." Alternatively, an accuracy standard may specify that a general project condition survey should be performed to meet an IHO "Special Order" TVU standard of ±0.8 ft.*

Total vertical uncertainty (TVU): Component of total propagated uncertainty (TPU) calculated in the vertical dimension. TVU is a one-dimensional quantity with all contributing vertical measurement uncertainties included.

$$TVU_{max}(d) = \sqrt{a^2 + (b \times d)^2}$$

Where

a represents that portion of the [uncertainty](#) that does not vary with the depth

b is a coefficient which represents that portion of the [uncertainty](#) that varies with the depth

d is the depth

MULTI BEAM STANDARDS

7.3 TABLE 1 - Minimum Bathymetry Standards for Safety of Navigation Hydrographic Surveys

To be read in conjunction with the full text set out in this document, m = metres, all [uncertainties](#) at 95% confidence level, * = Matrix Reference.

Reference	Criteria	Order 2	Order 1b	Order 1a	Special Order	Exclusive Order
Chapter 1	Area description (Generally)	Areas where a general description of the sea floor is considered adequate.	Areas where underkeel clearance is not considered to be an issue for the type of surface shipping expected to transit the area.	Areas where underkeel clearance is considered not to be critical but features of concern to surface shipping may exist.	Areas where underkeel clearance is critical	Areas where there is strict minimum underkeel clearance and manoeuvrability criteria
Section 2.6	Depth THU [m] + [% of Depth]	20 m + 10% of depth *Ba5, Bb2	5 m + 5% of depth *Ba8, Bb3	5 m + 5% of depth *Ba8, Bb3	2 m *Ba9	1 m *Ba10
Section 2.6 Section 3.2 Section 3.2.3	Depth TVU (a) [m] and (b)	a = 1.0 m b = 0.023 *Bc7, Bd4	a = 0.5 m b = 0.013 *Bc8, Bd6	a = 0.5 m b = 0.013 *Bc8, Bd6	a = 0.25 m b = 0.0075 *Bc10, Bd8	a = 0.15 m b = 0.0075 *Bc12, Bd8
Section 3.3	Feature Detection [m] or [% of Depth]	Not Specified	Not Specified	Cubic features > 2 m, in depths down to 40 m; 10% of depth beyond 40 m *Be5, Bf3 beyond 40m	Cubic features > 1 m *Be6	Cubic features > 0.5 m *Be9
Section 3.4	Feature Search [%]	Recommended but Not Required	Recommended but Not Required	100% *Bg9	100% *Bg9	200% *Bg12
Section 3.5	Bathymetric Coverage [%]	5% *Bh3	5% *Bh3	≤ 100% *≤ Bh9	100% *Bh9	200% *Bh12

$$TVU_{max}(d) = \sqrt{a^2 + (b \times d)^2}$$

HYDROGRAPHIC SURVEY TECHNICAL SPECIFICATIONS

MAJOR SPECIFICATIONS USED IN THE HYDROGRAPHY:	
USACE	EM 1110-2-1003 Manual
NOAA	Hydrographic specifications survey and deliverables
IHO (International Hydrographic Organization)	Standards for Hydrographic Surveys (S-44)

EM 1110-2-1003
30 Nov 13

Table 3-1. Recommended Depth Accuracy Standards for Corps of Engineers Surveys of Federal Navigation Projects based on Performance Test Results. ¹

Project	Typical ² Repeatability (feet)	Typical ³ Standard Deviation (± feet at 95%)
Coastal Deep Draft Projects (15>d<75 ft):		
Dredge measurement & payment surveys		
Channel clearance/acceptance		
Project condition surveys		
Maintenance Dredging (soft sand/silt bottom)	0.3 ft	±0.8 ft
New Work or Rock Cuts ⁴	0.2 ft	±0.8 ft
Coastal Shallow Draft Projects (d<15 ft)	0.3 ft	±0.8 ft
Inland Navigation Projects (d<15 ft)	0.3 ft	±0.5 ft

US Army Corps of Engineers. "EM 1110-2-1003 Hydrographic Surveying"
November 30, 2013

SINGLE BEAM VERSUS MULTIBEAM

Feature	Single-Beam Echo Sounder (SBES)	Multibeam Echo Sounder (MBES)
Coverage	Narrow – one depth point per ping	Wide swath – covers a fan-shaped area in one pass
Data Resolution	Low to medium	High – detailed seafloor morphology and 3D models
Survey Efficiency	Slower – requires multiple passes for large areas	Faster – large areas mapped in fewer passes
Best Use Cases	Shallow water, spot checks, budget-limited surveys	Engineering-grade surveys, dredging, hazard detection
Equipment Cost	Lower – simpler hardware	Higher – advanced sensors and motion compensation
Setup Complexity	Easier – minimal ancillary systems	Complex – requires frequent sound velocity profiling, patch testing, etc
Data Processing	Simple – basic depth profiles	Advanced – generates DTM, backscatter, point clouds
Accuracy	Good for single points; limited QC options	High accuracy with motion and latency corrections
Applications	Dual-frequency (fluid mud channels), beach monitoring profiles, quick condition surveys.	Full-bottom coverage for dredging, construction, habitat mapping

GEOPHYSICAL SURVEY

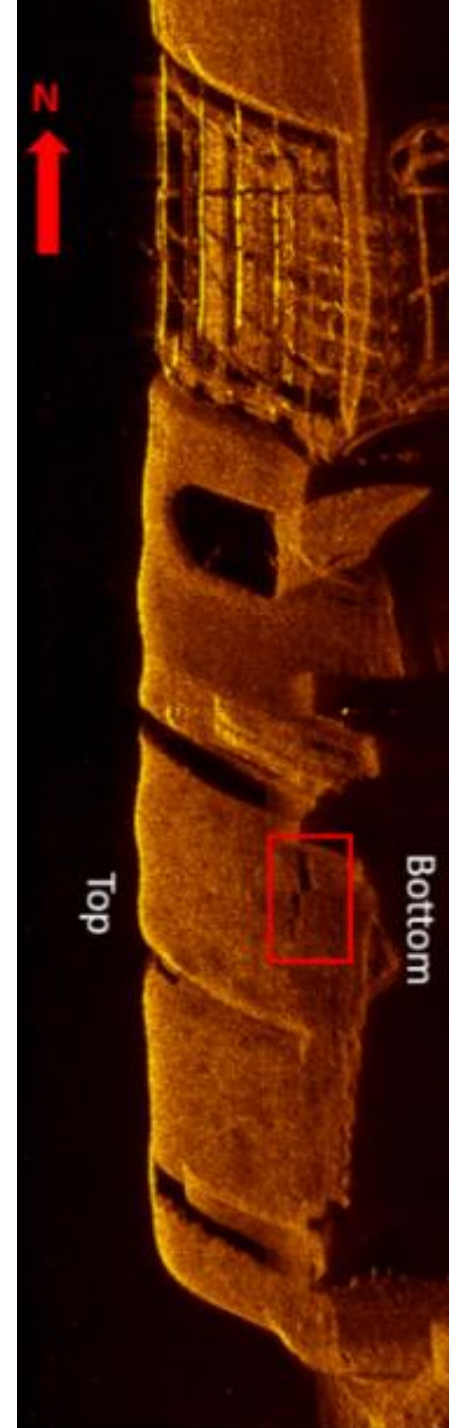
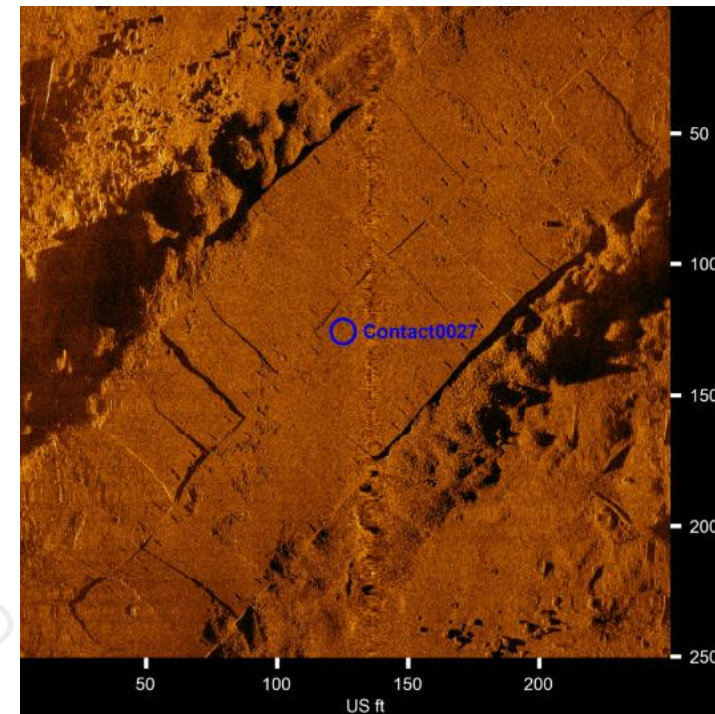
BRIEF OVERVIEW

BACKSCATTER:

measures how much acoustic energy returns to the sonar in decibels – seafloor characterization.

SIDE SCAN SONAR:

high resolution imaging tool - inspection, object detection, cultural resources



GEOPHYSICAL SURVEY

BRIEF OVERVIEW

MAGNETOMETER:

measures the magnetic field, which changes when approaching ferrous objects, called magnetic anomalies.

SUB-BOTTOM:

Seafloor penetration, seabed geology, pipeline detection.

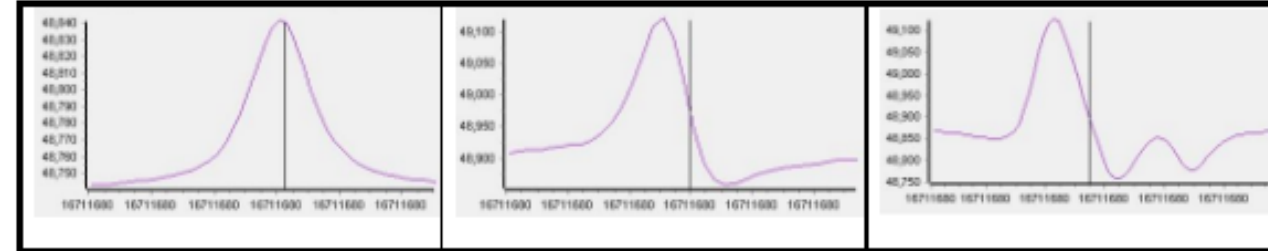
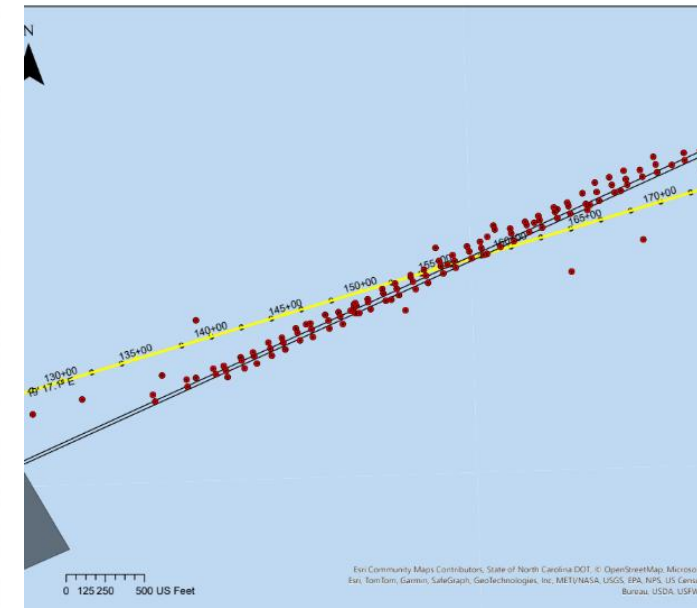
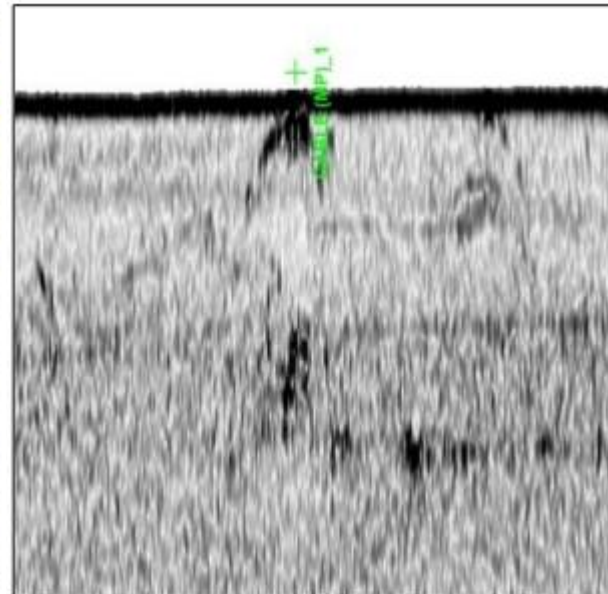
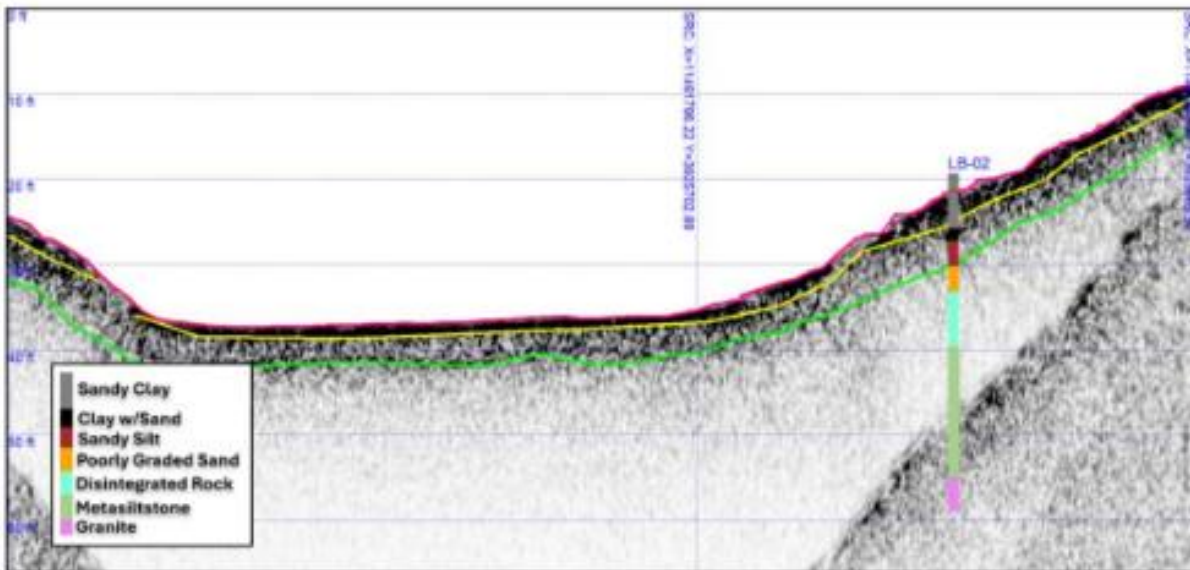
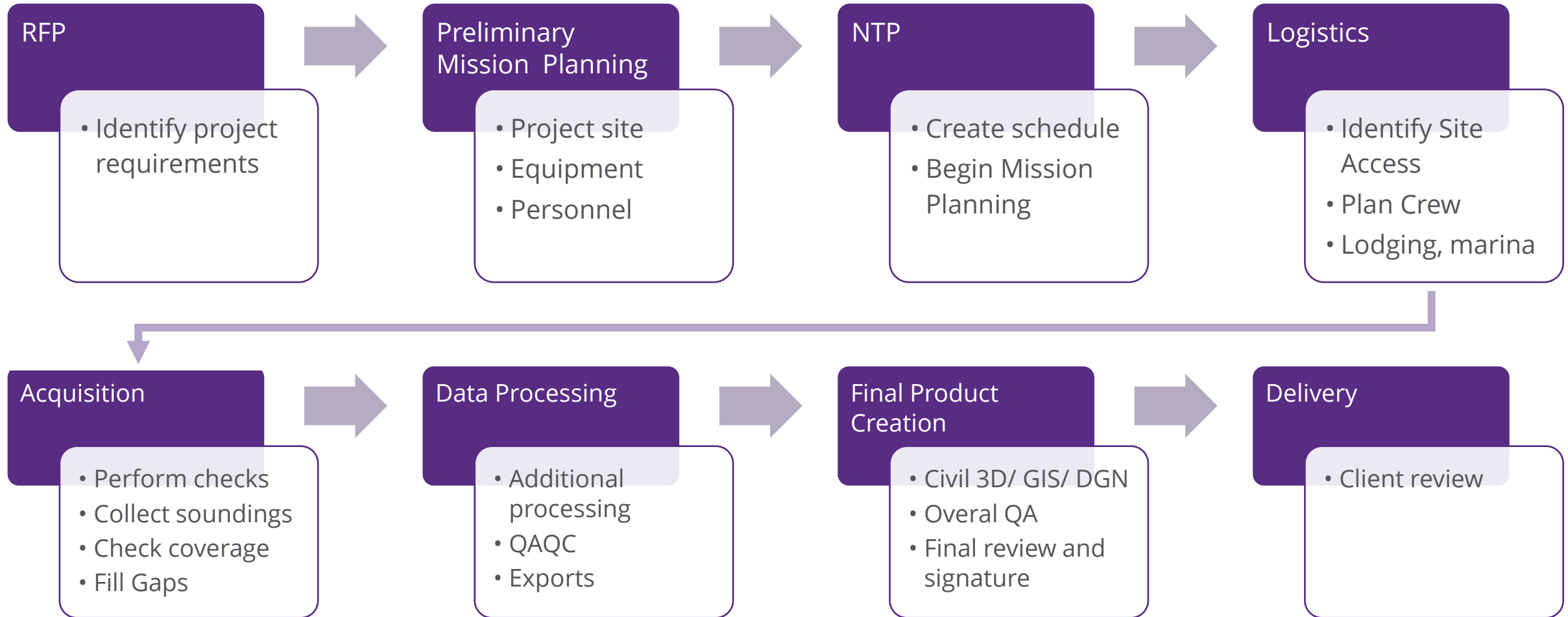


Figure 6. Examples of magnetic anomalies, from left to right: monopole, dipole, and complex anomalies.



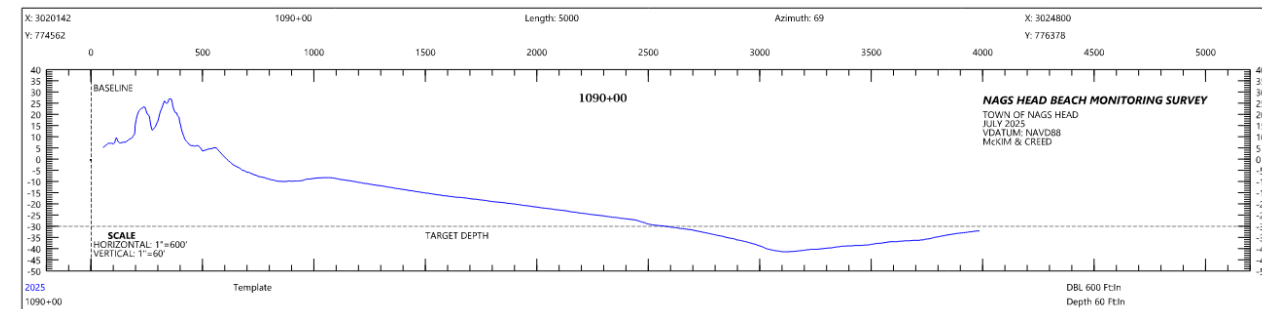
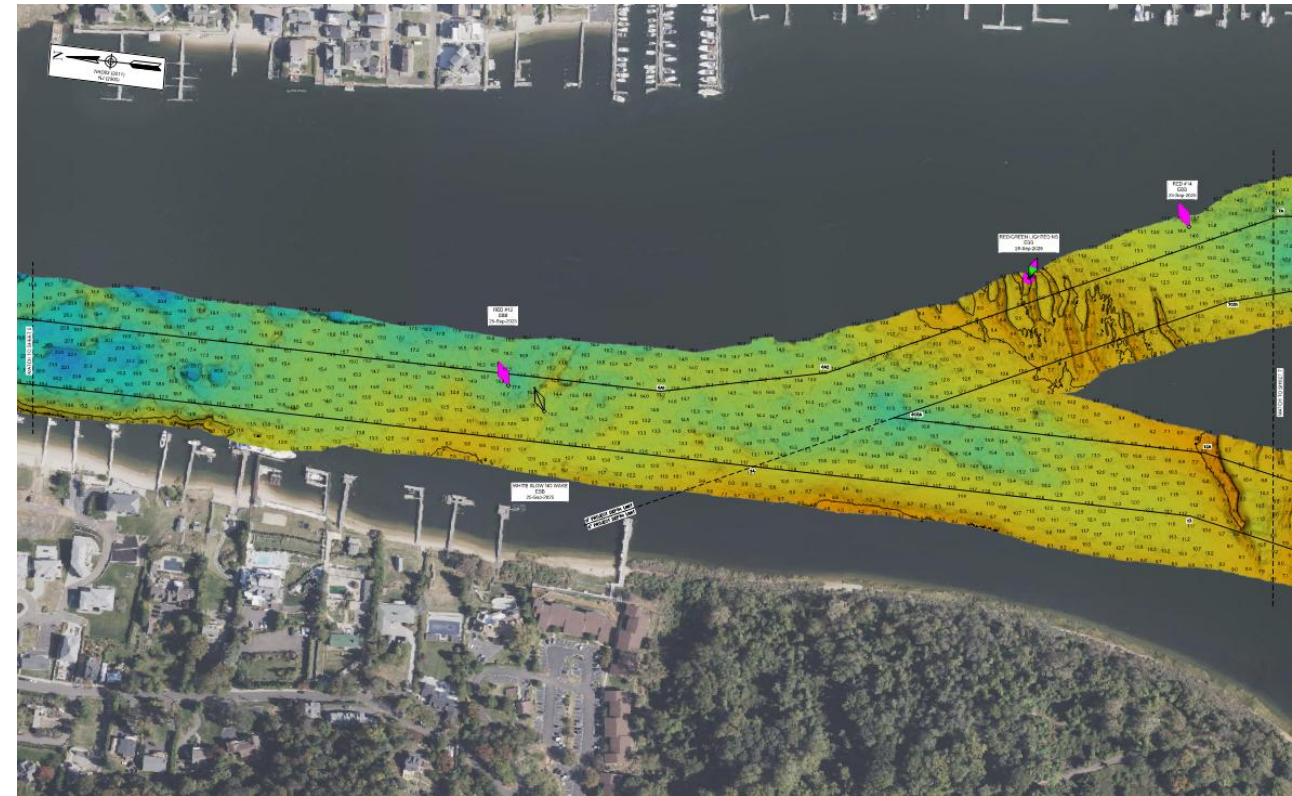
BATHYMETRIC PROJECT WORKFLOW



PROJECT CONSIDERATIONS

My client requested a bathymetric survey, what should I consider?

1. Client goals
2. Client related specifications
3. Does the survey require a PLS or CH for supervision or oversight?
4. Site access
5. Site-specific tides, depths, sound velocity, currents
6. Final products: requested vs needed
7. 100% bottom coverage vs profile survey
8. "Resolution" requirements:
 - Horizontal and Vertical
 - Object detection
 - Contour



PROJECT CONSIDERATIONS

Selecting the appropriate survey vessel

CREWED OR MANNED

Waves, weather, tides and currents

Water depth, distance from shore

Number of crew members

Type of Equipment



UNMANNED, REMOTE CONTROL, OR AUTONOMOUS

Size, cost, and functionality

Shore-based operation

Battery life, obstacle avoidance, recovery procedures



OTHER CONSIDERATIONS

Boat ramps or other access

Electric vessel requirements



SAFETY CONSIDERATIONS

HSE REQUIREMENTS

- Working near or on the water
- Specific training for safe vessel operations
- Site or client specific trainings (OSHA, Duke Power Safe)
- Additional safety equipment (PFD, life ring, distress signals, AIS system, VHF radio, radar, EPIRB (satellite distress alert beacon))

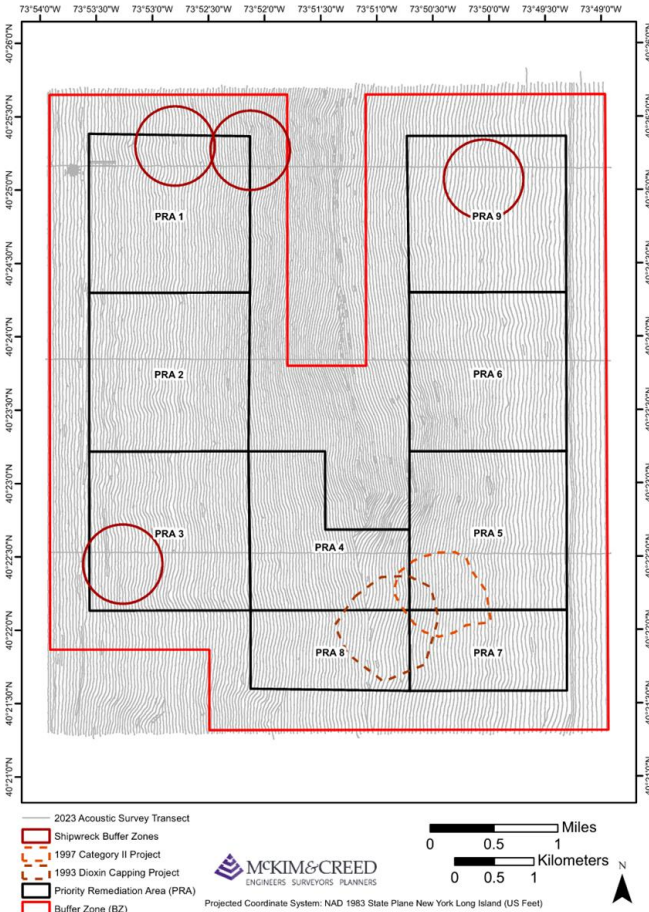
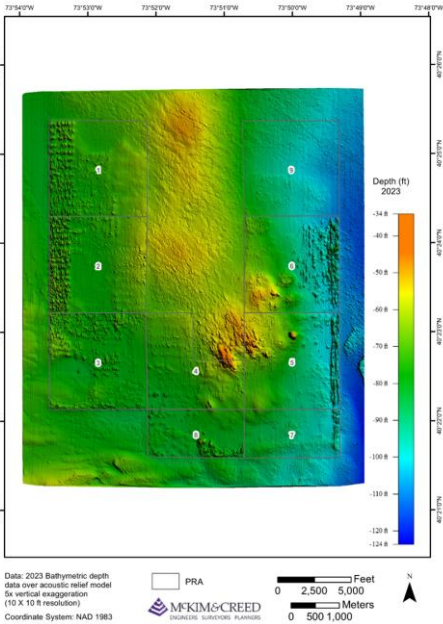


PRELIMINARY MISSION PLANNING

Contour requirements/ Line spacing to meet specifications and coverage requirements

Survey plan: Control, limits, access, geodesy, line creation.

635	26398 Feet	00:52:08
636	26398 Feet	00:52:08
637	26398 Feet	00:52:08
638	26398 Feet	00:52:08
639	26398 Feet	00:52:08
640	26398 Feet	00:52:08
641	26398 Feet	00:52:08
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647	26398 Feet	00:52:08
648	26398 Feet	00:52:08
649	26398 Feet	00:52:08
650	26398 Feet	00:52:08
651	26398 Feet	00:52:08
Total Lines: 651	17185012 Feet	565:39:32
Total transit time:		21:40:00
Total time:		587:19:32



PRELIMINARY MISSION PLANNING

SITE ANALYSIS

[Nautical Charts](#)

[EHydro](#)

Google Earth

Site 1
Site 2
Site 3



BATHYMETRIC DATUMS

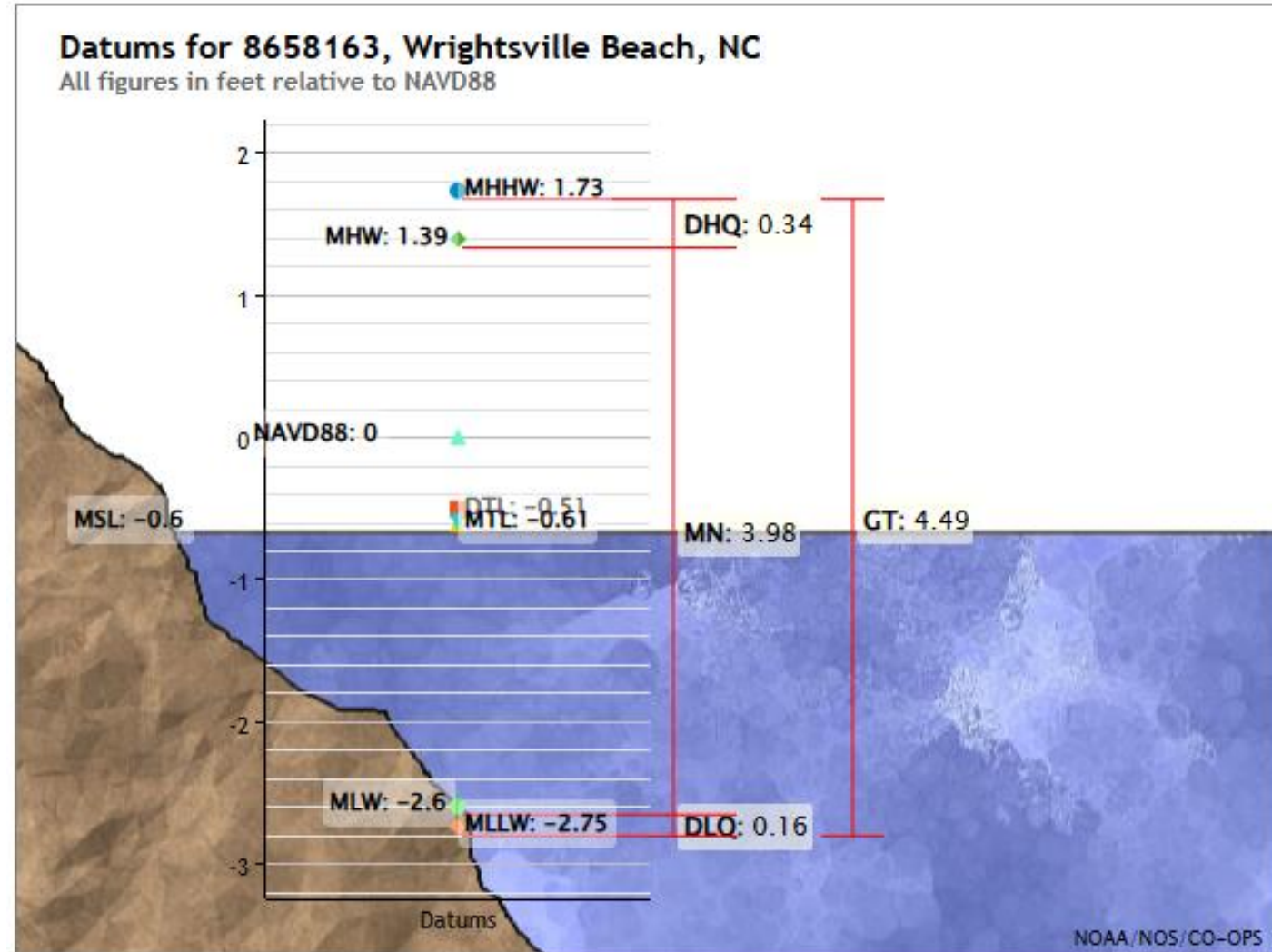
VERTICAL REFERENCING FOR BATHYMETRY CAN VARY GREATLY BY CLIENT, PROJECT, LOCATION, ETC.

Tidal datums are based on an epoch of 19 years (we are still using the 1983-2001 Epoch)

Tide Datums: MLLW, MLW, MLT, MHW, MHHW, Chart Datum, MSL, LAT (2 epochs).

Going from a vertical datum to a tidal datum

<https://vdatum.noaa.gov/vdatumweb/>





ONLINE VERTICAL DATUM TRANSFORMATION

INTEGRATING AMERICA'S ELEVATION DATA

- Home
- About VDatum
- Download
- Docs & Support
- Contact Us

Regional Information

* Region :

Horizontal Information

	Source	Target
Reference Frame:	NAD83(2011)	NAD83(2011)
Coord. System:	Geographic (Longitude, Latitude)	Geographic (Longitude, Latitude)
Unit:	meter (m)	meter (m)
Zone:		

☒ Vertical Information

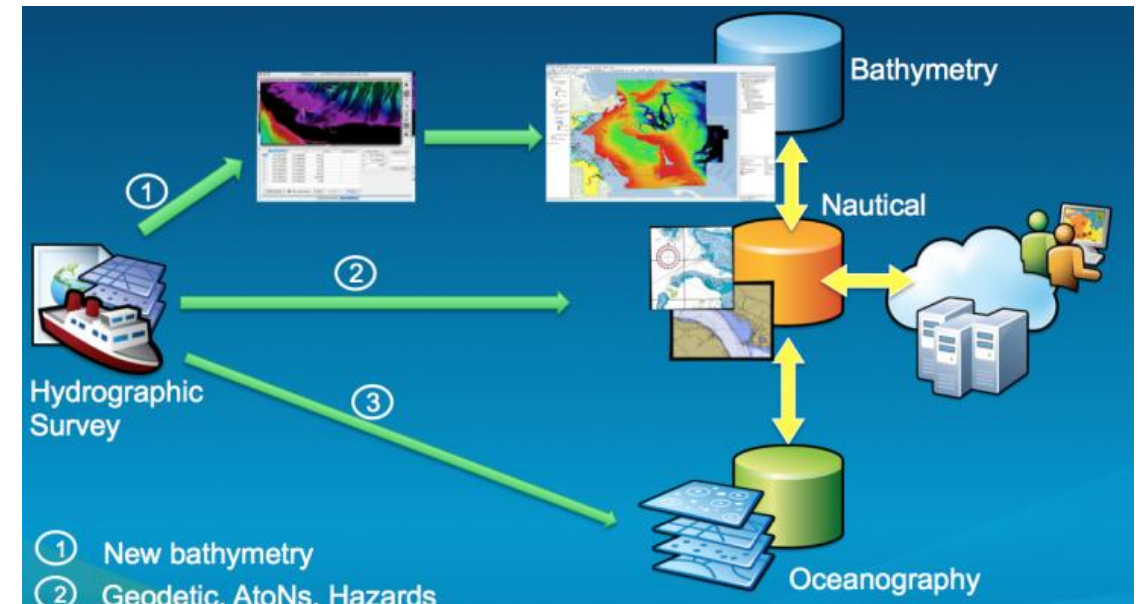
	Source	Target
Reference Frame:	NAVD 88	NAVD 88
Unit:	meter (m)	meter (m)
	☒ Height ☐ Sounding	☒ Height ☐ Sounding
	☐ GEOID model:	☐ GEOID model:

- Point Conversion
- ASCII File Conversion

Input		Output
Latitude: e.g. 33.7586 or 33 45 30.9600	Transform Reset DMS	Latitude:
Longitude: e.g. -118.7691 or -118 46 8.7600		Longitude:
Height: e.g. 3.037 Drive to on map Reset Map		Height: Drive to on map Reset Map
☐ to DMS		
Vertical Uncertainty: ☒ 1 sigma ☐ 95% Confidence		
☐ Add Observation Vertical Uncertainty		
Vertical Uncertainty (+/-):		
☒ Valid Tidal area ☒ Non-Tidal area ☒ Non-Valid area		
☒ CRD ☒ IGLD85 ☒ SVU area		

DATA PROCESSING

- Data Transfer
- Data Management
- Cloud-based processing: virtual, remote
- Quality Control checks
- Exports



DATA PROCESSING

MERGING BATHYMETRIC AND TERRESTRIAL DATA SETS

What to consider:

Different vertical datums

Datasets of varying resolutions

Different coverage

Most recent

Vertical uncertainty

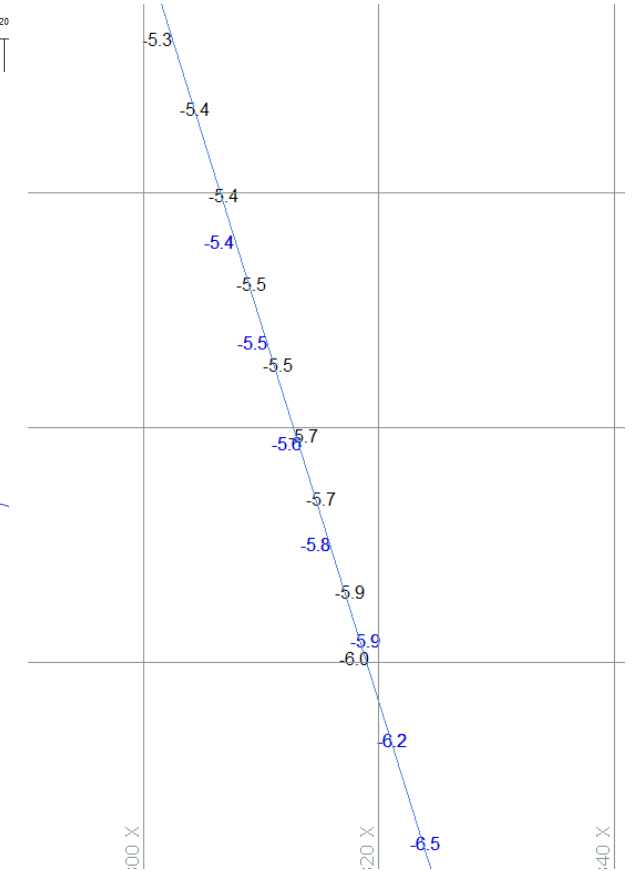
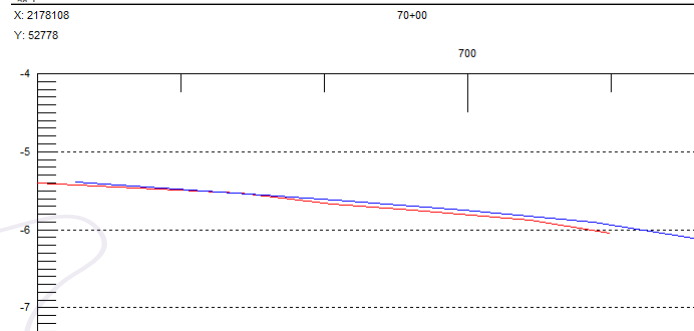
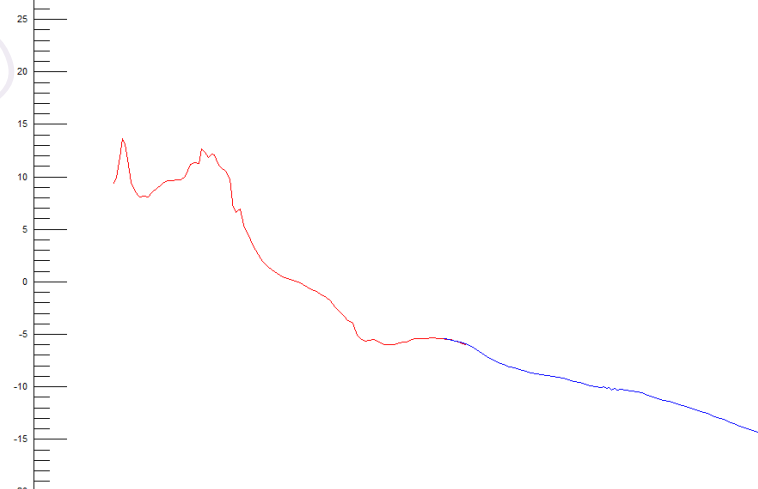
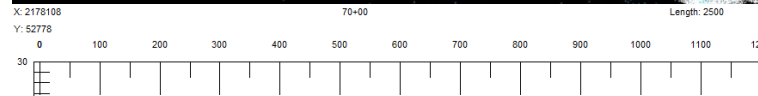
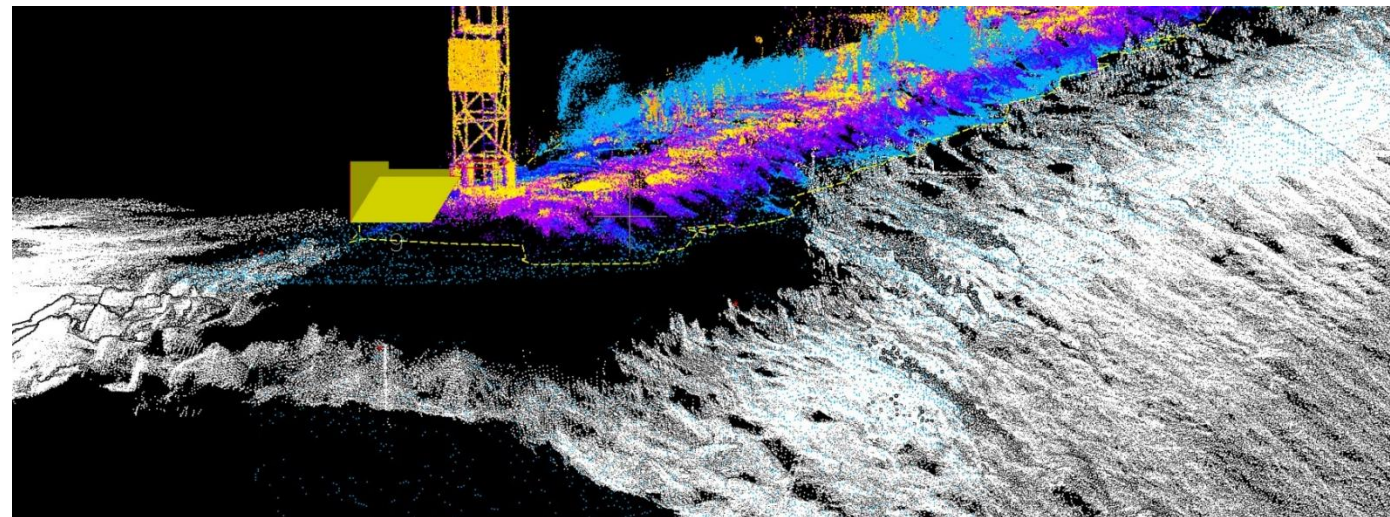
Comparing overlapping data:

Create cross-sections

Surface-to-surface comparison

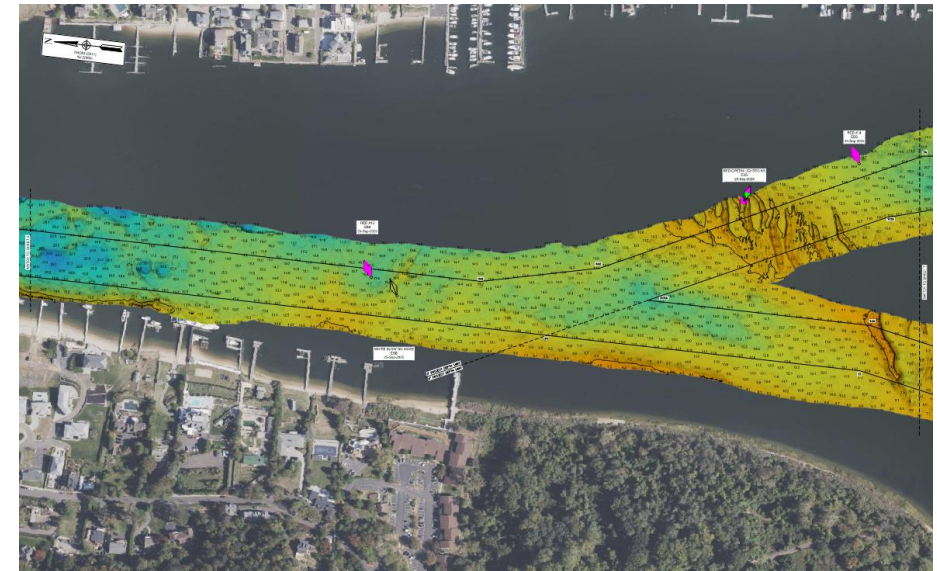
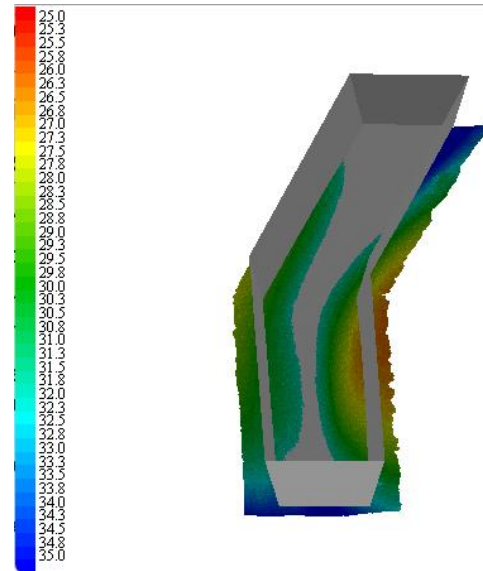
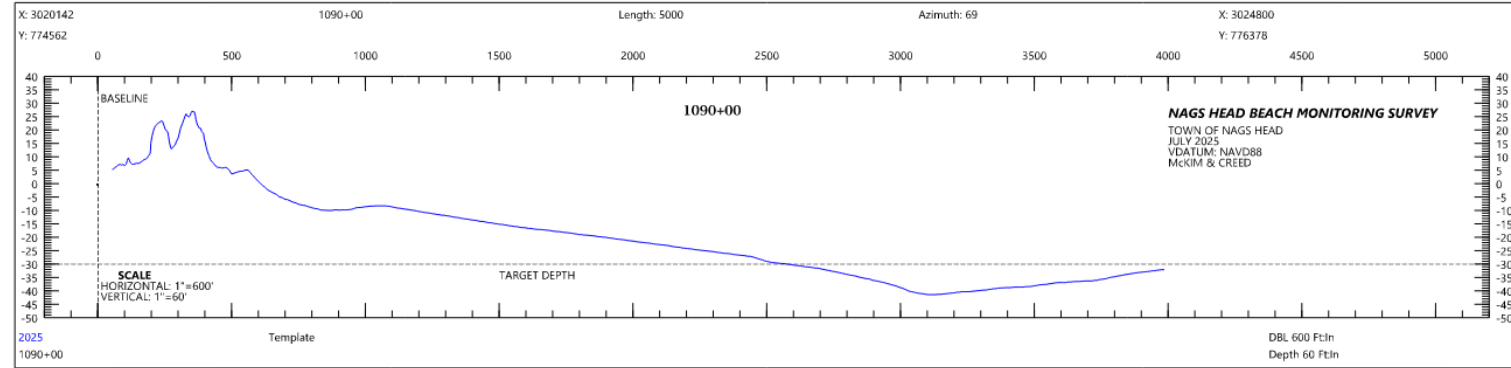
Plan view analysis

Spot checks



FINAL PRODUCTS

- Point files (ascii)
- Raw and edited files
- Target list/report
- Drawing (Civil 3D, DGN, GIS)
- Signed and seal map
- Cross-sections
- Volume Calculations
- Channel Condition Report
- Metadata
- Survey Report (calibrations, methods, analysis)



WANT TO LEARN MORE ABOUT BATHYMETRIC SURVEYING?

- Self study (books, online tutorials, manufacturer manuals, etc)
- On the job training
- THSOA.org – The Hydrographic Society of America
- US Hydro Conference – Wilmington 2027



Educational Programs through Colleges and Universities:

IHO.int – International Hydrographic Organization - [Recognized Hydrographic Programs](#)- Cat A and B:

- University of New Hampshire/NOAA-UNH Joint Hydrographic Center)
- University of Southern Mississippi (USM)
- University of New Brunswick

Other Programs:

- Marine Technology – CFCC



The program has **two certification levels**:

- **Level 1 - Certified Master Hydrographer (CMH)**
- **Level 2 - Certified Hydrographer (CH)**

[Obtain NSPS Certified Hydrographer Certification](#)



<https://digital.mckimcreed.com/hydrographic-survey-mckim-creed>

Thank you.
QUESTIONS

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